Forward Guidance and the Dynamics of Bank Credit

The Bank Balance-Sheet Channel of Monetary News

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January 2026

Abstract

This paper studies how forward guidance affects bank credit supply. Using high-frequency identification around FOMC announcements and local projections on U.S. bank-level data, I find a pronounced asymmetry: contractionary forward guidance triggers an immediate and persistent decline in lending, while expansionary guidance produces weak and delayed loan growth. The mechanism operates through bank balance sheets: tightening shocks reduce equity and raise leverage, pushing banks toward binding capital constraints and forcing credit contraction. A quantitative model with bank constraints matches these dynamics and implies limited stimulus from easing guidance.

* I would like to extend my gratitude to Fabio Ghironi, Phillip Bond, Chang-Jin Kim, and Kemal Ozhan for their feedback and support. This paper benefited from comments by Jeffrey Campbell, Refet Gürkaynak and participants in the T2M and RES conferences. I'd also like to thank participants in the MTI workshop at UW.

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1 Introduction

According to the Federal Reserve Board's website, forward guidance "is a tool that central banks use to tell the public about the likely future course of monetary policy." [1] The Federal Reserve uses forward guidance to shape the expectations of the public at large, individuals and firms alike, so that they have reliable information on which to base their investment decisions. As of 2013, other central banks, such as the Bank of Japan, Bank of England, and European Central Bank, have added forward guidance into their set of monetary policy instruments, with mixed results.[2] Since the latter half of the 1990s, forward guidance has been incorporated into FOMC announcements. As central banks continue to make use of this policy mechanism, it's important to continue filling in the gaps in the literature surrounding the effects of this policy tool, which is the intention of this project.

The focus of this paper is on analyzing the effects of forward guidance on bank credit, specifically looking at how banks adjust credit issuance in response to both contractionary and expansionary forward guidance shocks. A very large amount of work already exists on the effects of innovations to the federal funds rate on bank balance sheets and risk-appetite, such as those by Bernanke and Gertler (1995), Adrian and Shin (2010), and Bruno and Shin (2015).

However, there has yet to be any work completed that specifically looks at the effects of the Federal Reserve's forward guidance policy on these variables, operating through an expectations channel, as opposed to an immediate rate change. Therefore, my findings contribute to the literature on the bank balance-sheet channel of monetary policy by illustrating how forward guidance uniquely shapes the timing and composition of bank credit adjustments.

In order to identify forward guidance shocks, I employ an established methodology in the forward guidance literature that originated in Gürkaynak et al. (2005) [3], where the authors gather high-frequency price data for federal funds rate futures from a thirty-minute window surrounding FOMC meetings, spanning the years 1995 to 2020. The purpose of tracking

[1] https://www.federalreserve.gov/faqs/what-is-forward-guidance-how-is-it-used-in-the-federal-reserve-monetary-policy.htm

[2] Gertler (2017) discusses the initial limited effects of forward guidance in Japan, identifying the BOJ's lack of credibility as a contributing factor to the relative ineffectiveness of its forward guidance efforts.

[3] From here on referred to as GSS (2005)

the data within such a narrow daily window is to isolate the effects that purely extend from FOMC announcements on these asset prices, as opposed to other macroeconomic news that might also influence the prices. At this point, the next step of the procedure is to use the tools of PCA to extract two factors from the effects of FOMC announcements on asset prices: a so-called "target" and a "path factor." They are then rotated so that the latter has zero correlation with the former. The path factor is our so-called "forward guidance shock," while the target factor is the shock to the current federal funds rate.

Bauer and Swanson (2023) demonstrate that, despite the high-frequency nature of the federal funds futures data, it is still correlated with macreconomic news. In order to resolve this issue, I follow their procedure of regressing each realization of the path factor on contemporaneous realizations of macroeconomic news. I then collect the residuals of this regression and sum the derived bi-quarterly forward guidance shocks and divide them by two for an average measure each quarter.

In a similar spirit to that of Campbell et al. (2012) and Bundick and Smith (2020), who assess the effects of this shock on real GDP, the GDP Deflator, and real investment, I extend the analysis to explore the impact of forward guidance shocks on banks' portfolio reallocation and leverage dynamics.

I find that, on impact, bank leverage rises in the wake of a contractionary forward guidance shock, which is owing to the fact that banks' equity falls faster than their asset holdings. I also find that lending falls immediately and, in line with Van den Heuvel et al. (2002), the adjustment of C&I loans is sluggish, relative to other loans categories, and consumer loans is a major driver of the change in total lending, which displays a large and immediate adjustment. Additionally, in line with the results of Bernanke and Gertler (1995), Kashyap et al. (1996) and Kashyap and Stein (2000), who study the federal funds rate's effects on bank balance sheets, I find that following a contractionary forward guidance shock, the total loan volume issued by commercial banks steadily decreases over time.[4]

Importantly, my results show an asymmetry in the responses of both bank leverage and lending in response to forward guidance shocks, depending on whether the shock is

[4]Bernanke and Blinder (1992) point out that the slow and gradual response of loans can be attributed to their contractual nature.

expansionary or contractionary. The responses of these variables to a contractionary signal from the central bank are larger than their responses to an expansionary shock. Not only this, but the dynamics are different. At early horizons, the response of bank balance sheets to a signal of monetary tightening is immediate. Under an expansion, by contrast, bank leverage and credit adjust gradually, peaking at much later horizons than what is observed in a contractionary environment. I argue that this is owing to capital requirements that force banks to change their lending behaviors in a manner unique to contractionary environments, providing a bank balance-sheet channel for monetary policy.

Taken together, the findings suggest that forward guidance can powerfully contract bank credit, but it has difficulty expanding or stimulating it. Given the fact that the primary interest in forward guidance is its potential to stimulate the economy at the zero bound, these findings indicate that forward guidance to be least effective in the precise state of affairs where it is supposed to be most relevant.

Related Literature

This paper aims to fill a key gap in both the forward guidance literature as well as the "conventional"[5] monetary policy literature by assessing the effects of contractionary forward guidance shocks on bank balance sheets and holdings of risky assets. Presently, there exists a large body of literature analyzing the effects of shocks to the federal funds rate on bank balance sheet adjustments, like those of Bernanke and Blinder (1992), Van den Heuvel et al. (2002), and Kashyap and Stein (2000), who evaluate the effects of shocks to the federal funds rate on banks. Adrian and Shin (2010) explored the effects of financial markets on bank balance sheets and how banks adjust leverage in response to appreciations in their assets. Gambacorta and Shin (2018) show evidence of a differential impact of federal funds rate shocks on bank lending behavior that depends on bank capitalization. Although this literature investigates the effects of monetary policy on bank balance sheets, it lacks an analysis of the effects of forward guidance on these factors in particular.

[5]The distinction between the Federal Reserve's adjustment of its current federal funds rate target and the Federal Reserve's forward guidance policy is often framed as a distinction between "conventional" and "unconventional" monetary policy. From here on, I follow Benigno (2025) in not referring to forward guidance as unconventional.

In terms of the forward guidance literature, there is a diverse range of analysis that has been conducted, with some authors such as Campbell et al. (2012), Nakamura and Steinsson (2018), and Bundick and Smith (2020) looking at macro aggregates like GDP, unemployment, inflation, consumption, and investment. Other papers break down the responses of these variables even further, like Kroner (2021), who looks at how differences in firm-level uncertainty result in heterogeneous effects of forward guidance on investment. Several papers have investigated the effects of forward guidance on key asset prices such as treasuries and corporate bonds, along with commodity prices and the S&P 500, as in Gürkaynak et al. (2005), and Bauer and Swanson (2023). Swanson (2021) also looks at the effects of forward guidance on dollar/euro and dollar/yen exchange rates. However, just as the monetary policy shock literature, which looks at balance sheet effects, lacks an investigation of the effects of forward guidance, the forward guidance literature lacks an investigation of the effects on bank balance sheets. The aim here is to bridge this gap.

The methodology employed in this paper to identify forward guidance shocks follows GSS 2005 and its subsequent applications in related papers Gürkaynak (2005), Swanson and Jayawickrema (2023), and Bauer and Swanson (2023). GSS 2005 builds upon a single-factor approach to measuring the impact of monetary policy surprises on asset prices developed by Kuttner (2001), and Cochrane and Piazzesi (2002), where GSS 2005 shows an improvement upon the single-factor approach via their two-factor approach.

Ultimately, this study contributes to the broader literature examining the intricate linkages between monetary policy shifts and the financial sector. By focusing on forward guidance shocks, this paper sheds light on how these shocks influence portfolio allocations of commercial banks, a critical channel through which monetary policy propagates to the real economy. Now that forward guidance appears to have become a permanent fixture of the Federal Reserve monetary policy apparatus, understanding these dynamics is becoming highly relevant. My findings provide new empirical evidence on banks' portfolio reallocation behavior and highlight the nuanced ways in which banks balance trade-offs between liquidity, risk, and profitability across different horizons. These insights stress the importance of forward guidance as a tool for influencing financial intermediation and, ultimately, macroeconomic outcomes.

In the next section, section 2, I will provide a more detailed explanation of how the forward guidance shocks are constructed and identified. Then, in section 3, I will describe the variables involved in my analysis of bank behavior in response to these shocks. Section 4 contains the empirical results of this paper. Section 5 will setup the quantitative model for our analysis. Section 6 will discuss the results of the quantitative model and the final section, section 7, will conclude.

2 Forward Guidance Shocks

The construction of the forward guidance shocks follows the methodology employed in GSS 2005. To begin, I obtain high-frequency data from Acosta et al. (2024) that spans July 1995 to July 2020. The data is collected from a 30-minute window surrounding every FOMC announcement in that time-span where the window starts 10 minutes before and ends 20 minutes after the announcement time. I end up with a total of 204 observations. Within those 30-minute windows, changes in five key futures contracts are tracked: the current-month and three-month-ahead federal funds futures contracts, and the second, third, and fourth eurodollar futures contracts, which have an average of 1.5, 2.5, and 3.5 quarters to expiration, respectively. Please see the appendix for additional detail on how these changes are constructed, which simply follows the methodology employed by Kuttner (2001) and GSS 2005.[6]

Using principal components analysis, two factors are extracted. The factors are rotated so that the first is correlated with changes to the current federal funds rate and the second has no loading on current changes in the federal funds rate. The first is called the target factor, and the latter is called the path factor. The path factor is the forward guidance shock.

Figure 1 is a plot of the forward guidance shocks that spans the years 2012 to 2024. The red bars represent monetary easing surprises and the blue bars monetary tightening surprises. The height of the bars represent the magnitude of the surprise. In accordance

[6]Some outlier observations, such as the emergency meeting after 9/11, were omitted. Please see the appendix for more details.

with what one might expect, the magnitude of the surprise upon the advent of COVID-19 in the beginning of 2020 is huge, with much more easing than previously anticipated. This illustrates why observations after 2019 are omitted from this analysis.

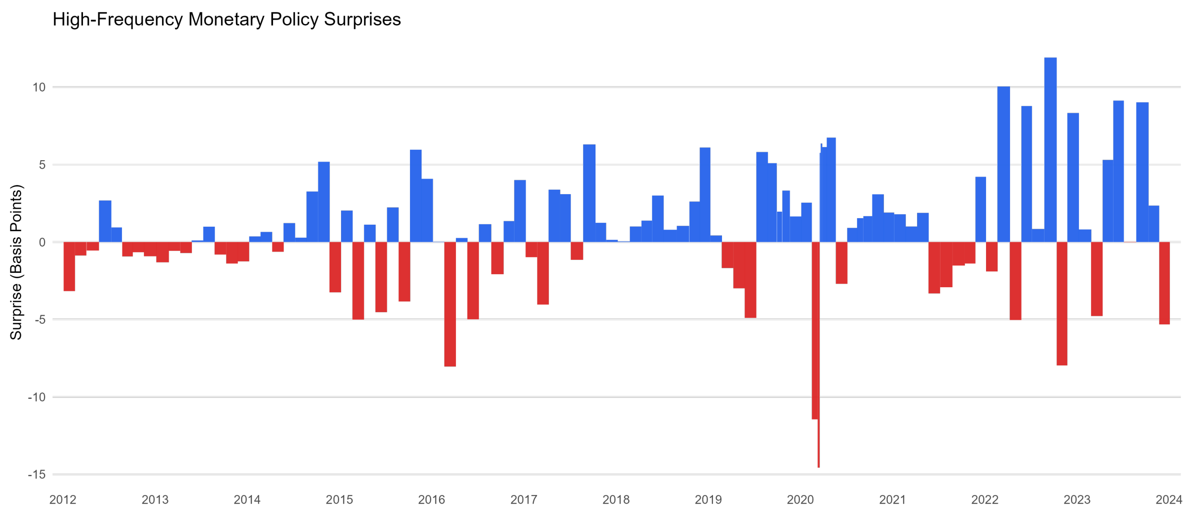


Figure 1: Forward Guidance Shocks Over Time

This plot was retrieved from the San Francisco Fed's website. The data is part of the Center for Monetary Research.

Much of the bank-level data is only available at quarterly-level frequencies, whereas the forward guidance shocks extracted from PCA are bi-quarterly. To reconcile this frequency mismatch, I sum up the forward guidance shocks that I previously obtained and divide that quantity by two for an average quarterly measure. Before summing the shocks, and in step with Cieslak (2018), Kroner (2021), and Bauer and Swanson (2023), I take each forward guidance shock and regress it on a vector of macroeconomic and financial variables data which were made available before the FOMC announcement. This is to control for the effects that macro and financial news might have on the estimation of both the forward guidance shock and the banks' adjustments to their balance sheets.

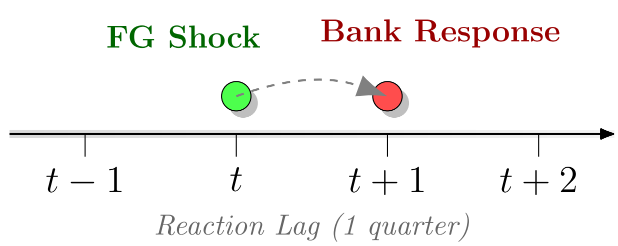
Following Bauer and Swanson (2023), I use six variables that are related to the Federal Reserve's monetary policy decisions: nonfarm payrolls surprise, employment growth, S&P 500, yield curve slope, commodity prices, and treasury skewness. The nonfarm payrolls surprise is calculated as the difference between the value of the most recent nonfarm payrolls release and the expectation for that release based upon surveys of financial market participants. Employment growth is calculated in terms of the log difference between the current and previous years' nonfarm payroll employment releases, which follows Cieslak (2018). The S&P 500 variable represents a log difference in the stock market index the day prior to the announcement and 65 trading days prior to the announcement. The yield curve slope is the difference between the slope of the yield curve the day before the announcement and three months prior to the announcement. For commodity prices, I look at the log difference in the Bloomberg Commodity Spot Price index between three months prior to the announcement and the day prior to the announcement. Last, the treasury skewness measure is the implied skewness of the 10-year Treasury yield, following Bauer and Chernov (2024). I add the residuals of each bi-quarterly regression to a single quarterly shock that represents the final forward guidance shock. The news-effects are projected out in the following manner:

$$Path_{\tau} = \alpha + \beta V_{\tau} + u_{\tau}$$

Where $Path_{\tau}$ is one of the bi-quarterly forward guidance shocks (path factor) at time τ, and V_{τ} is our vector of controls for macroeconomic news that is released prior to the FOMC announcement, as described above. Then, summing our bi-quarterly shock to a single measure we obtain:

$$\psi_t = \frac{\sum_{\tau \in t} u_{\tau}}{2}$$

Due to the delay in banks' ability to respond to these shocks, I offset the shocks by a single quarter so that I estimate banks' reactions to the forward guidance shock one quarter after the initial forward guidance shock is observed. This procedure also follows Kroner (2021) and Bauer and Swanson (2023), as illustrated below:



The appendix provides details on the macroeconomic and financial variables used to control for the effects of news releases on surprises to the futures data. As a validation exercise, Appendix A.2 Figure 12 reports the responses of real GDP, real investment, and the GDP deflator to contractionary and expansionary forward-guidance shocks. The aggregate responses are broadly consistent with the standard forward-guidance literature.

3 Bank Data

When examining the effects of forward guidance shocks on bank balance sheets, I'm primarily interested in looking at broad trends in the banking sector. My dataset covers a sample spanning from 1995Q3 to 2024Q3. Data on many of the bank balance sheet items are publicly available through the Board of Governors of the Federal Reserve System and the Federal Deposit Insurance Corporation (FDIC). However, a few of the key variables of interest in my study are not available from these sources over the desired date range. Those variables are bank loans and bank leverage. Loan data is obtained from FFIEC Call Reports via Wharton Research Data Services (WRDS).

In order to obtain a broad measure of bank leverage, I pull data on its individual components and combine them to make a single measure. The data for the leverage components only extend to the year 2020 in the Board of Governors' public database. To get a measure that extends through 2024, I construct a broad measure of leverage via data from WRDS; specifically, I obtain the necessary data from Compustat's Capital IQ bank fundamentals. The Compustat data contains information on about 10,800 large and mid-sized U.S. commercial banks that serve as the representative sample of "commercial banks." For the remaining bank-level variables in my dataset, the Board of Governors and FDIC provide public data over the appropriate date range.

To ameliorate any concerns regarding this approach, I acquired data from the Board of Governors' public database and constructed a measure of leverage from 1995 to 2020 which is similar to my leverage measure constructed via the Compustat data. Then I compared the magnitudes and directions of the impulse responses of the leverage measures from both data sources over this mutually shared horizon (1995-2020). The magnitudes and directions of the IRFs for leverage, assets, debt, and equity were virtually identical across both datasets.

Hence, my 1995 to 2024 Compustat sample is a consistent representation of broader banking industry leverage trends.

In constructing a measure of leverage for the banking system at large, I follow Adrian et al. (2014) in employing the following specification:

$$\text{Leverage}_t = \frac{\text{Assets}_t}{\text{Equity}_t}$$

which is the inverse of the He et al. (2017) capital ratio. Data on these banks' holdings of U.S. Treasury securities was taken from Call Reports in Compustat. All of the remaining data in my study is publicly available from either the FDIC or the Board of Governors' websites.

4 Forward Guidance Shocks and Bank Balance Sheets

4.1 Local Projections: Baseline Results

For several bank-level variables, such as: total loans, assets, equity, etc., the data are recorded in billions of dollars. Due to scale mismatch with my forward guidance shocks, I apply a log-difference transformation to convert levels into growth rates, following Galí and Gertler (1999).

This transformation helps mitigate the scale mismatch, allowing us to interpret the coefficients as the percentage change in the variable in response to a shock. To estimate the dynamic responses of bank-level variables to a contractionary forward guidance shock, I use the Local Projections (LP) method from Jordà (2005). Unlike vector autoregressions (VARs), LPs estimate the response at each horizon separately, which has an enhanced capacity for handling nonlinearities and time variation. Additionally, we have an identified exogenous shock being fed into the model, which means a VAR would not be appropriate here. As a result, my impulse response function is specified as:

$$\Delta_\tau \log(y_{t+\tau}) = \mu_i^\tau + \alpha^\tau FG_{i,t}^{\text{con}} + \gamma^\tau FG_{i,t}^{\text{exp}} + \sum_{i=1}^{4} \gamma_i^{\tau\prime} Z_{t-i} + \varepsilon_{t+\tau}, \quad \tau = 0, 1, \ldots, 20,$$

where $\Delta_\tau \log(y_{t+\tau}) \equiv \log(y_{t+h}) - \log(y_{t-1})$ is the percentage change in the variable at horizon τ, $FG^{\text{con}}_{i,t}$ represents the contractionary forward guidance shock, and $FG^{\text{exp}}_{i,t}$ is its expansionary counterpart. They are constructed as follows:

$$FG^{\text{con}}_{i,t} = \max(\text{shock}_t, 0)$$
$$FG^{\text{exp}}_{i,t} = \max(-\text{shock}_t, 0)$$

With "shock$_t$" being the residual of the regression that projects out the effects of macro news. Z_t is a vector of control variables, which includes macro aggregates in the form of log transformations of real GDP, the GDP deflator, real investment, the federal funds rate. It also includes bank-level controls, such as deposits/assets, risk-weighted assets as a percentage of total assets, AOCI/equity, and total assets. μ_i represents bank-level fixed effects. The coefficients α and γ measure the impact of a one standard deviation contractionary and expansionary forward guidance shock, respectively, on the growth rate of y_τ (in percentage points).

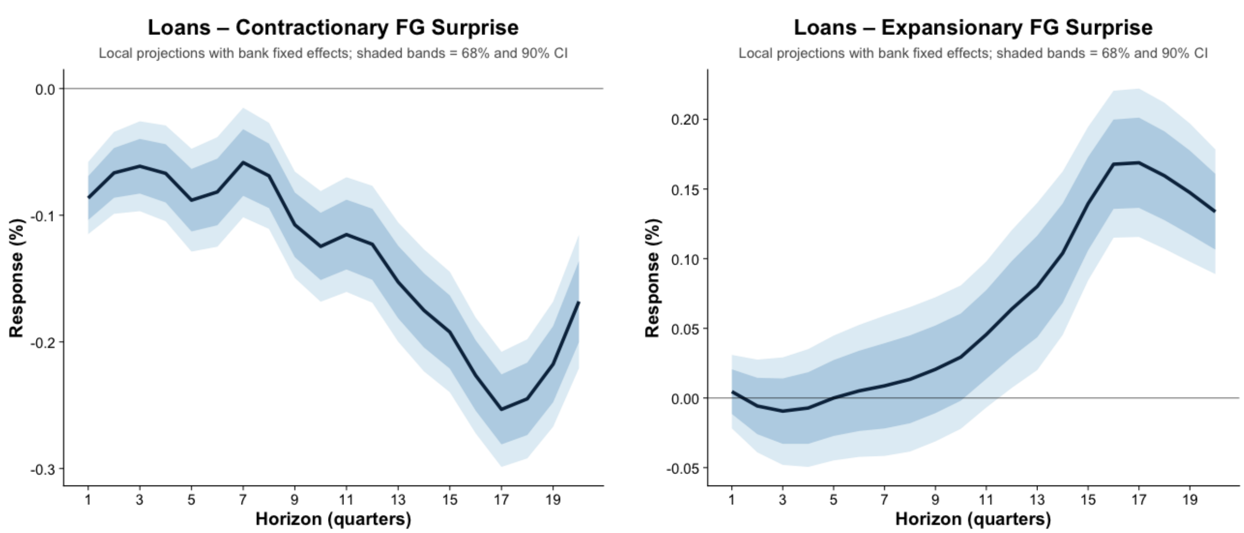


Figure 2: Loan Responses to Forward Guidance Shocks

The figures show the impulse responses of total bank loans to a one-standard-deviation contractionary forward-guidance shock on the left and expansionary forward-guidance shock on the right. The inner dark band corresponds to the 68% confidence band, while the outer light band corresponds to the 90% confidence band.

Figure 2 shows the impulse responses of total loans to contractionary and expansionary forward guidance shocks over 20 quarters.

For each of the IRFs, I conduct a Wald test in order to test the null hypothesis that the IRF responses across the 20 horizons are jointly zero. For the IRFs presented in Figure 2, the null hypothesis is strongly rejected, indicating statistically significant responses over time. I also looked at the signifiance at each horizon. The pointwise estimates are significant at the 5% level across every horizon for loans following a contractionary surprise, with the exception of horizons three and seven. For loans following an expansionary surprise, the pointwise estimates become significant at the 13th horizon and remain significant over the remainder of the horizons thereafter.

The left panel of Figure 2 shows an immediate and significant decline in lending following a contractionary signal from the Fed. The credit contraction sustains itself and even intensifies until 17 quarters out from the initial shock, at which point lending begins its rebound. From this we can conclude that, in an environment of monetary tightening, banks react quickly and forcefully to forward guidance surprises. The reaction is not only immediate, but it is also sustained over a long period of time.

For the right panel, by contrast, the first 12 point estimates are insignificant at the 5% level. This means that the change in lending is indistinguishable from zero until 13 quarters after the initial shock. Once, we reach quarter 13 and beyond, we see a statisically significant increase in lending. It is also noteworthy that the absolute value of the lending response is smaller in an expansionary environment than it is in a contractionary environment. With a trough of about 29 basis points by quarter 17 in the case of tightening, and a peak of about 19 basis points in the case of an expansion.

An asymmetry emerges in this figure: forward guidance shocks have more of an impact on bank lending in a contractionary environment than they do in an expansionary environment. The tightening results in a fast, large, and significant contraction in lending, which is sustained over several years. The expansion on the other hand, doesn't have a tangible effect on lending until several quarters after the initial surprise. The reaction in this latter case is both delayed and smaller.

Another feature is that the change in lending accelerates over time in both cases. Along the initial horizons of the left panel, we see an immediate drop in lending, then lending sharply decelerates, indicating a lag in the response of bank lending in a contractionary environment. Bernanke (1992) and Kashyap et al. (1996) point out that banks have contractual obligations with other firms that they cannot immediately terminate. Additionally, even when the contractionary shock hits, previously approved loans will still fund at the previously agreed upon rate. During the approval process, a rate may get locked-in before the shock, and finalize well after the shock. Hence, subsequent to the FOMC announcement, many loans are still being finalized at older rates.

A second factor involved with the deceleration in lending has to do with "relationship lending." After a contractionary credit shock, some banks are better able to smooth loans rates than others, especially banks funded through deposits with inelastic rates, as shown in Berlin and Mester (1999). A third factor relates to demand-side preexisting commitments where, as a result of investments and expenditures by firms having been planned for several months leading up to the FOMC announcement, firms continue with short-term borrowing for things like inventory finance; a trend observed by Gertler and Gilchrist (1994).

Additionally, a bank balance sheet metric of relevance to credit movements is leverage. The details for how the leverage variable is constructed can be found in section three. There are conflicting findings in the literature regarding the response of bank leverage to macroeconomic shocks, that is, whether bank leverage is procyclical or countercyclical. Brunnermeier and Pedersen (2009), Adrian and Boyarchenko (2012), Adrian et al. (2013), and Adrian, Etula, and Muir (2014) document results consistent with procyclical broker-dealer leverage. On the other hand, He and Krishnamurthy (2013), Di Tella (2017), and He, Kelly, and Manella (2017) find that intermediary leverage is countercyclical.

A similar asymmetric pattern emerges when looking at the dynamics of leverage in Figure 3, depending upon whether one examines its response to a contractionary or expansionary forward guidance shock. Both IRFs exhibit joint significance at the 1% level over the 20 quarter window. The pointwise estimates for the contractionary panel are highly significant at horizons 1, 3, and 13. For the expansionary panel, the pointwise estimates are significant at the 5% level at horizons 8 and 11.

The left panel of Figure 3 displays an immediate and significant increase in leverage for banks over the first several quarters in the wake of a contractionary forward guidance shock. The leverage variable is constructed as assets/equity. Both total assets and equity fall in the immediate aftermath of the shock, but equity falls faster, resulting in an overall increase in leverage, creating balance sheet pressure for banks upon the impact of the shock.

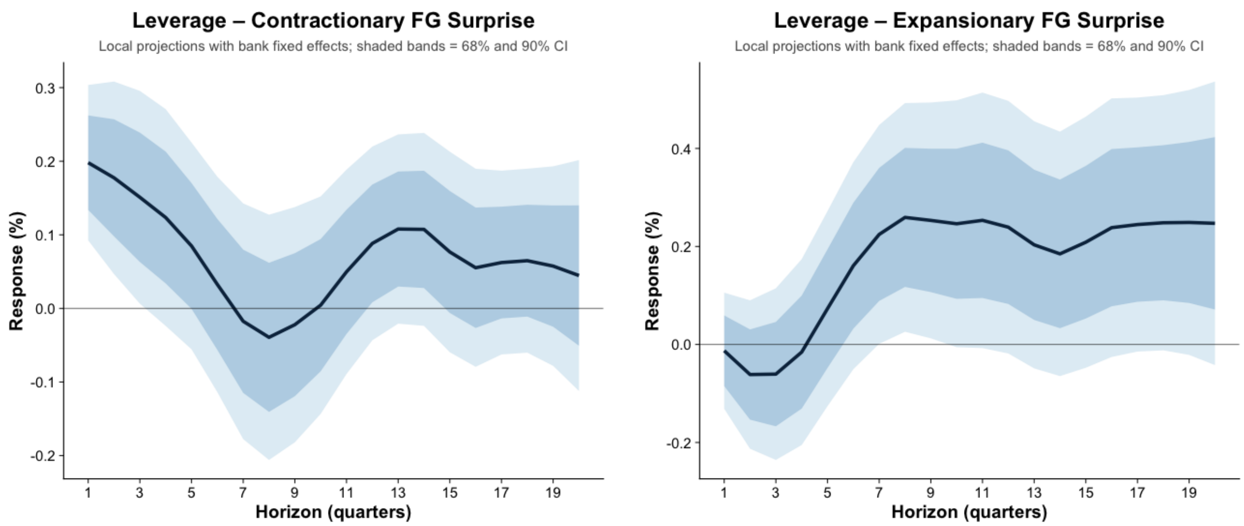


Figure 3: Leverage Responses to Forward Guidance Shocks

The figures show the impulse responses of bank leverage to a one-standard-deviation contractionary forward-guidance shock on the left and expansionary forward-guidance shock on the right. The inner dashed band corresponds to the 68% confidence band, while the outer solid band corresponds to the 90% confidence band.

On the other hand, when the economy experiences monetary easing, there is no significant effect on leverage over the first eight quarters, where it begins to increase, this indicates that a forward guidance signal of monetary easing takes time to translate into any discernible effect on bank leverage. This mirrors the asymmetric pattern that emerges in bank lending: we have an immediate and significant reaction in a contractionary environment that creates a jump in bank leverage on impact, and a delayed, back-loaded reaction of bank leverage to expansionary forward guidance surprises.

The countercyclical pattern observed in bank leverage can be understood through the framework outlined by He et al. (2017). The dynamics of leverage fluctuations depend on whether financial intermediaries operate under "equity constraints" or "debt constraints." Different types of institutions tend to fall into one category or the other. For instance, hedge funds, which rely heavily on borrowing, are typically debt constrained, whereas commercial banks (the focus of the present study) are more often equity constrained. In times of economic contraction, these differences in constraints lead to opposite leverage responses. When hedge funds experience tighter borrowing limits, they are forced to reduce leverage by selling assets, which often end up in the hands of commercial banks. As a result, leverage moves in opposite directions for these two groups of intermediaries.

Equity-constrained financial institutions, as modeled in studies such as Bernanke and Gertler (1989), Holmstrom and Tirole (1997), and Brunnermeier and Sannikov (2014), see their equity capital decline in a downturn, which reduces their overall risk-bearing capacity. Although they may respond by cutting back on debt financing, the reduction in equity capital is typically larger than the decrease in debt, leading to an overall rise in leverage. This dynamic explains why leverage tends to be countercyclical for these institutions.

By contrast, models developed by Brunnermeier and Pedersen (2009), Adrian et al. (2014), and Adrian et al. (2013) suggest that intermediaries facing strict debt constraints exhibit procyclical leverage. Hedge funds, for example, often operate under borrowing limits that tighten during economic downturns, forcing them to liquidate assets in order to meet margin requirements. This process of deleveraging is so strong that it outweighs the drop in equity, causing their leverage to decline alongside the broader economy. Since these institutions offload assets rapidly, equilibrium prices fall, further reinforcing the procyclical pattern of leverage adjustments.[7]

4.2 Local Projections: Capitalization Channel

The baseline results indicate that there is an asymmetry in commercial banks' lending depending on whether the forward guidance shock is expansionary or contractionary. The leverage IRFs also contain an asymmetry that points us in the direction a capitalization channel that lies behind the asymmetry in lending. Specifically, leverage rises in the wake of a FG contraction, but moves insignificantly in an expansionary environment. Since capital constraints only bind from below, changes in capitalization can only force bank adjustments when it falls toward the constraint.

[7] He, Kelly, Manela (2017), 23.

To this end, I test whether the loan response to forward-guidance shocks depends on banks' proximity to regulatory capital constraints. I sort banks into different "bins" based on their Tier 1 capital ratio. Then I re-estimate the local projections by interacting forward-guidance surprises with lagged Tier 1 capital-buffer bins. Capital buffer bins are delineated using each bank's lagged Tier 1 capital ratio relative to a 6% regulatory constraint. Specifically, I measure the lagged capital buffer as the difference between the bank's Tier 1 ratio and 6%.

$$\text{Buffer}_{i,t-1} = 100\left(\text{Tier1}_{i,t-1} - 0.06\right),$$

Banks are then assigned to their respective mutually exclusive bins depending on whether they are below the requirement, between 0 and 1 percentage points above the requirement, between 1 and 2 percentage points above the requirement, or between 2 and 4 percentage points above the requirement. Banks with Tier 1 buffers of at least 4 percentage points serve as the omitted reference group. I notate these categories by the following indicators:

$$D^{\text{below}}_{i,t-1} = \mathbf{1}\{\text{Buffer}_{i,t-1} < 0\}, \qquad D^{0\text{–}1}_{i,t-1} = \mathbf{1}\{0 \le \text{Buffer}_{i,t-1} < 1\},$$

$$D^{1\text{–}2}_{i,t-1} = \mathbf{1}\{1 \le \text{Buffer}_{i,t-1} < 2\}, \qquad D^{2\text{–}4}_{i,t-1} = \mathbf{1}\{2 \le \text{Buffer}_{i,t-1} < 4\}.$$

The resulting IRFs show how lending responds to forward-guidance shocks by proximity to the regulatory capital constraint.

For each horizon h, I estimate

$$\begin{aligned}\Delta_\tau \log(y_{t+\tau}) = \alpha_i^{(h)} + \lambda_t^{(h)} + \sum_{g\in\mathcal{G}} \beta_{g,c}^{(h)}\left(FG_t^{+} D^{g}_{i,t-1}\right) + \sum_{g\in\mathcal{G}} \beta_{g,e}^{(h)}\left(FG_t^{-} D^{g}_{i,t-1}\right) \\ + \theta^{(h)} Z_{i,t-1} + \psi_c^{(h)}\left(FG_t^{+} Z_{i,t-1}\right) + \psi_e^{(h)}\left(FG_t^{-} Z_{i,t-1}\right) + u_{it}^{(h)}.\end{aligned}$$

Where, $\Delta_\tau \log(y_{t+\tau})$ represents $\log(y_{t+h}) - \log(y_t)$ with the outcome variable being lending. Now, $\alpha_i^{(h)}$ denotes bank fixed effects and $\lambda_t^{(h)}$ denotes quarter fixed effects. $D^{g}_{i,t-1}$ represents the indicator for buffer-bin group g and bank i, with $g \in \mathcal{G} = [\text{below}, \text{0-1}, \text{1-2}, \text{2-4}]$. The vector $Z_{i,t-1}$ contains lagged bank-level controls:

$$Z_{i,t-1} = \left[\Delta y_{i,t-1},\ \frac{Deposits_{i,t-1}}{Assets_{i,t-1}},\ \frac{RWA_{i,t-1}}{Assets_{i,t-1}},\ \frac{AOCI_{i,t-1}}{Equity_{i,t-1}},\ Tier1_{i,t-1},\ 100\log(Assets_{i,t-1})\right]',$$

where

$$\Delta y_{i,t-1} = y_{i,t-1} - y_{i,t-2}.$$

If capital channel is relevant, it should be the case that as banks approach the capital constraint in the wake of a contractionary FG shock, they tighten lending more sharply. The IRFs below plot this interaction effect, with the coefficient of interest being $\beta^h_{g,c}$.

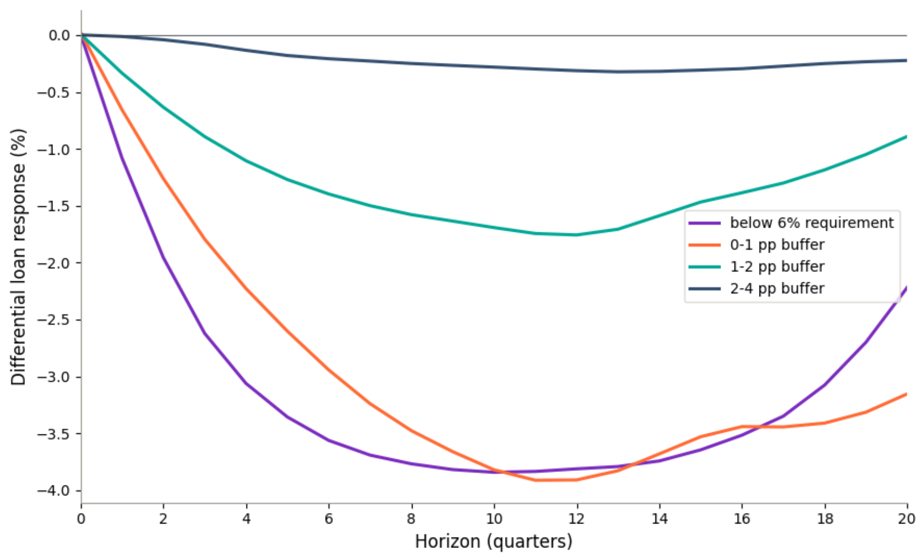


Figure 4: Lending Response to Contractionary FG: Sorted by Bank Capitalization

Figure 4 shows the interaction effect of contractionary forward guidance shocks with the banks' Tier 1 capital ratio. As anticipated, relative to the baseline level of capitalization ($\geq$ 4 pp buffer), banks contract credit more strongly as they come closer to the capitalization constraint upon realization of the contractionary shock. The absolute magnitude of contraction is an increasing function of capital tightness.

The average effects of an expansion is found in the appendix. The interaction effect is statistically indistinguishable from zero at every horizon, preserving the asymmetry under this more robust specification. The error bands around each IRF in Figure 4 indicate that the response of lending at each buffer bin is significant at all horizons. This figure is also displayed in the appendix.

4.3 Securities Valuation Losses and the Leverage Response

To examine whether the leverage response is associated with valuation losses on banks' securities portfolios, I construct a measure of unrealized gains and losses on each bank's total securities book. The measure is defined as the difference between the reported fair value and amortized cost of held-to-maturity securities, plus the difference between the reported fair value and amortized cost of available-for-sale securities. Let $FV_{i,t}^{HTM}$ and $AC_{i,t}^{HTM}$ denote the fair value and amortized cost of held-to-maturity securities for bank i in quarter t, and let $FV_{i,t}^{AFS}$ and $AC_{i,t}^{AFS}$ denote the corresponding values for available-for-sale securities. I define the total unrealized securities gain/loss as

$$UGL_{i,t}^{SEC} = \left(FV_{i,t}^{HTM} - AC_{i,t}^{HTM}\right) + \left(FV_{i,t}^{AFS} - AC_{i,t}^{AFS}\right).$$

I scale this total valuation wedge by bank equity, so that the outcome captures the change in unrealized securities losses relative to the equity base relevant for leverage.

$$y_{i,t}^{SEC} = 100 \times \frac{UGL_{i,t}^{SEC}}{Equity_{i,t}}.$$

Negative values of $y_{i,t}^{SEC}$ indicate that the fair value of the bank's securities portfolio is below amortized cost, corresponding to unrealized valuation losses.

Because banks differ in how they classify securities, I do not require a bank to report both held-to-maturity and available-for-sale positions in a given quarter. Instead, I construct the valuation wedge separately for each accounting category and then aggregate across categories when the relevant fair-value and amortized-cost combination is available.

I estimate the response of this valuation wedge using the same local-projection framework as in the lending and leverage specifications. For each horizon ($h = 1, \ldots, 20$), the dependent variable is the change in the equity-scaled securities valuation wedge between quarter t and quarter $t+h$. The regression includes the contractionary and expansionary forward-guidance shock measures, bank fixed effects, and the same lagged macroeconomic controls used in the baseline specifications. I also control for lagged bank size, lagged equity-to-assets, and lagged securities exposure. Standard errors are clustered at the bank level.

Figure 17 reports the response of total securities valuation losses to a contractionary forward guidance shock. The response is negative, and significantly so, at all horizons in the window. This is indicative that contractionary forward guidance impacts bank equity through its securities portfolio quite substantially — the valuation wedge falls by roughly 0.15 to 0.35 percentage points of equity over the response horizon.

The results serve as evidence of a contributing mechanism, not a complete decomposition of the leverage response. Leverage may also respond to changes in loan-loss expectations and other market equity valuations. Within the category of securities, it is also not clear which securities in particular are driving the results. As shows in the appendix, the fall in the value of commercial bank treasury holdings is a very minor element of the overall response of bank securities. The same is true of commercial bank MBS holdings. I defer the investigation of what particular securities primarily drive the overall change in total bank securities after the FG shock to another study.

4.4 Cross-Sectional Analysis

The IRFs above display a pattern — substantial reductions in lending in wake of a contractionary shock, paired with a jump in bank leverage; while expansionary shocks produce virtually no change in lending until later horizons. I argue that a sufficient condition for this behavior and a major driving factor is the interaction with capital requirements and bank balance sheets. Basel III requires a minimum Tier-1 capital ratio of 6%. When banks are pushed up against this threshold, they have no choice but do de-lever, which will force them to contract credit issuance.

Given the asymmetric movements in leverage, it follows that contractionary shocks push some banks closer to the capital threshold, which forces behaviors in banks that we do not see in an expansionary environment. I argue that we have the following chain of effects: the Fed surprises markets with an announcement about the future path of interest rates, market participants react more strongly to bad news than to good news and this shows up in asset prices. Bank equity falls strongly after a contractionary, but rises mildly after an expansionary shock. This creates the asymmetric patterns in leverage and, therefore, credit.

To test this hypothesis I run a few tests. Figure 13 displays, as predicted, the asymmetric responses of bank shareholder equity using the same LP framework as above. A strong and statistically significant drop in shareholder equity follows a contractionary forward guidance shock, while an expansionary shock produces little to no response in equity.

The next thing I look at is how sensitive lending is to the capital ratio — do changes in regulatory capital predict changes in lending? I run the following regression:

$$\Delta \log(\text{Loans}_{i,t}) = \alpha_i + \delta_t + \beta \, \text{Tier1Ratio}_{i,t-1} + \Gamma' Z_{i,t-1} + \varepsilon_{i,t},$$

Where α_i is bank fixed effects and δ_t is quarter fixed effects. Standard errors $\varepsilon_{i,t}$ are clustered at the bank level. $Z_{i,t-1}$ is a vector of controls. Table 2 establishes a positive relationship between bank capitalization and subsequent loan growth. In bank and quarter fixed-effects specifications, a higher Tier 1 capital ratio is associated with faster growth in lending, controlling for bank size, deposit funding, risk-weighted asset intensity, and loan-performance measures. The estimated coefficient on the lagged Tier 1 ratio is positive and statistically significant in the baseline specification ($\beta = 0.21$, s.e. $= 0.075$), and remains essentially unchanged when controlling for AOCI relative to assets ($\beta = 0.210$, s.e. $= 0.075$). When I also include equity-to-assets as a further balance-sheet control, the Tier 1 estimate is attenuated but remains positive and statistically significant ($\beta = 0.184$, s.e. $= 0.082$). Overall, these results provide evidence that bank capital positions predict lending dynamics in the cross section, which is consistent with the view that capital constraints are indeed relevant for credit supply. Next, I test whether the capitalization–lending relationship becomes substantially stronger when banks are close to binding regulatory capital requirements.

Table 3 shows the results of a test that determines whether regulatory capital constraints amplify the sensitivity of lending to capitalization. I estimate a piecewise specification that allows the slope of lending in response to bank capitalization to differ depending on whether a bank is above or below a regulatory capital threshold. Specifically, I estimate

$$\Delta \log(\text{Loans}_{i,t}) = \alpha_i + \delta_t + \beta_{\text{below}} \, d^{-}_{i,t-1} + \beta_{\text{above}} \, d^{+}_{i,t-1} + \Gamma' Z_{i,t-1} + \varepsilon_{i,t},$$

where the dependent variable is quarterly loan growth, $\Delta \log(\text{Loans}_{i,t}) \equiv \log(\text{Loans}_{i,t}) - \log(\text{Loans}_{i,t-1})$, α_i denotes bank fixed effects, and δ_t denotes quarter fixed effects.

The key regressors are constructed from the bank's Tier 1 capital ratio relative to a regulatory threshold τ. Define Tier 1 distance to the requirement:

$$d_{i,t-1} \equiv \text{Tier1Ratio}_{i,t-1} - \tau.$$

Where $\tau = 0.06$. I then decompose this distance into its negative and positive parts:

$$d^-_{i,t-1} \equiv \max\{-d_{i,t-1}, 0\}, \qquad d^+_{i,t-1} \equiv \max\{d_{i,t-1}, 0\}.$$

The coefficient β_{below} captures the sensitivity of lending to capitalization when banks operate at or below the requirement , while β_{above} captures the corresponding sensitivity when banks hold capital buffers above the requirement. In this regression, if my hypothesis about the asymmetries laid out above is correct, then we'll have:

$$|\beta_{\text{below}}| > |\beta_{\text{above}}|,$$

Meaning that lending growth becomes substantially more sensitive to Tier-1 capitalization when banks are near the regulatory constraints. Table 3 shows $\beta_{\text{below}} = -1.888$ and $\beta_{\text{above}} = 0.209$, which is consistent with the proposed capital-constraint-channel mechanism.

The controls $Z_{i,t-1}$ include the same lagged bank characteristics as the previous regression. Standard errors are clustered at the bank level.

The table shows that the relationship between bank capitalization and lending is positive: as banks approach the capital constraint, they contract credit. The independent variable $\text{LowCapital}_{i,t-1}(\text{Tier } 1 \leq \text{p25})$ is an indicator variable for whether a particular bank i is in the bottom 25th percentile of capitalization in the sample at time $t-1$. I find a statistically significant and negative relationship between this indicator and the dependent variable: banks contract credit by about 4.7% when their capitalization falls to this level (SE: 1.7%). The interaction term tells us that as these banks' capitalization improves by one standard deviation, they expand credit by 0.01*(0.210+0.434) $\approx$ 0.64%.

Overall then, Table 3 provides evidence that lending becomes much more sensitive to capitalization when banks operate close to regulatory requirements.

Table 4 tests whether the lending response to forward-guidance (FG) surprises varies systematically with banks' Tier 1 capital buffers. I separate banks into various "bins" based upon their buffer at any point in time. The specification is as follows:

$$\Delta \log(\text{Loans}_{i,t}) = \alpha_i + \delta_t + \sum_{b \in \mathcal{B}} \beta_b \Big(FG_t^{\text{contr}} \times \mathbf{1}\{\text{BufferBin}_{i,t-1} = b\} \Big) + \Gamma' Z_{i,t-1} + \varepsilon_{i,t},$$

$$\Delta \log(\text{Loans}_{i,t}) = \alpha_i + \delta_t + \sum_{b \in \mathcal{B}} \gamma_b \Big(FG_t^{\text{exp}} \times \mathbf{1}\{\text{BufferBin}_{i,t-1} = b\} \Big) + \Gamma' Z_{i,t-1} + \varepsilon_{i,t}.$$

The dependent variable, $\Delta \log(\text{Loans}_{i,t})$, is quarterly loan growth for bank i in quarter t. The term α_i denotes bank fixed effects, which absorb time-invariant differences across banks, and δ_t denotes quarter fixed effects, which absorb aggregate conditions affecting all banks in a given quarter.

The key regressors interact the aggregate FG shock in quarter t with indicator variables for the bank's Tier 1 *capital buffer bin* in the previous quarter. Let $\mathcal{B}$ denote the set of buffer bins: below requirement, $[0, 1)$pp, $[1, 2)$pp, and $[2, 4)$pp; with ($\geq$ 4pp) used as the reference group. For each bank-quarter, $\mathbf{1}\{\text{BufferBin}_{i,t-1} = b\}$ equals one if bank i was in bin b at $t-1$, and zero otherwise.

One buffer bin b_0 is designated as the reference group and is not interacted with the FG shock in the summation. In this regression, the reference group is the highest-buffer bin (banks with at least 4 percentage points of Tier 1 capital buffer above the regulatory requirement). As a result, each coefficient β_b (or γ_b) measures how the sensitivity of lending to FG shocks differs for banks in buffer bin b relative to high-buffer banks.

With quarter fixed effects, this specification identifies differences in FG sensitivity across capital-buffer bins. Each coefficient β_b measures the incremental FG sensitivity for banks in buffer bin b, relative to the sensitivity of the highest-buffer bin. Once again, I split the regressions into contractionary FG_t^{contr} versus expansionary shocks FG_t^{exp}. Lagged bank controls are the same as before and standard errors are clustered at the bank level.

Table 4 presents evidence that the credit response to forward-guidance (FG) shocks does in fact depend on banks' Tier-1 capital ratios. For contractionary FG surprises, the lending response is state-dependent: banks operating below the Tier-1 requirement exhibit the largest decline in lending, with an estimated sensitivity of -0.007 (s.e.$= 0.002$). As banks' capitalization improves, their sensitivity to contractionary FG becomes gradually smaller. On the other hand, expansionary FG shocks generate fairly weak lending responses across buffer bins, with coefficients that are not only small in magnitude, but largely statistically insignificant (with the exception of a modest effect in the $[1, 2)$pp bin). Overall then, Table 4 implies that contractionary FG shocks reduce bank lending primarily when banks have limited capital buffers, consistent with a capital-constraint channel in which credit supply contracts most strongly when regulatory capital requirements are close to binding, and there is an asymmetry here between contractionary versus expansionary shocks.

5 Model with a Banking Sector

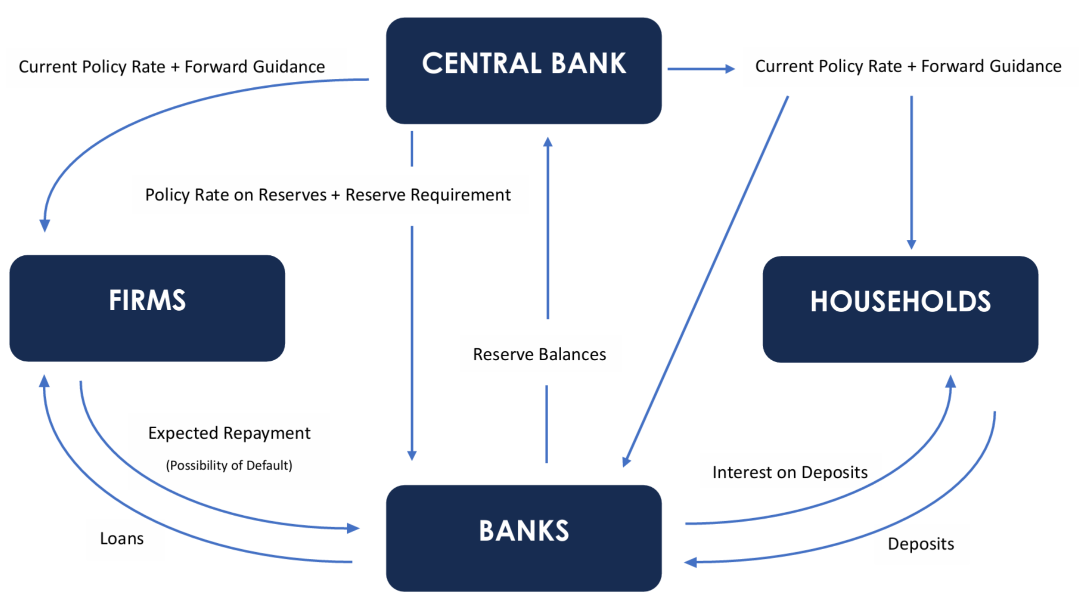


Figure 5: Model Structure

To interpret the empirical findings of the previous section, I construct a quantitative model that incorporates both a banking sector and forward guidance shocks. The basic structure of banking sector is borrowed from Benigno and Benigno (2021). In my banking sector, however, I incorporate an adjustment cost for loans. The banking sector is interconnected with the production side of the economy via intermediate-goods firms' financing of their inputs through loans, which they repay the banks with interest. There is an endogenously determined probability that the firms will not repay the loans, meaning they default.

The central bank uses the interest rate on reserves in order to control inflation. The interest rate on deposits and risk-free bonds are what drives changes in the household's consumption and saving patterns. Deposits and treasury notes are two instruments that the household can use for liquidity services. The model also includes various standard frictions, such as Calvo price-stickiness. The formulation of the forward guidance shock follows Campbell et al. (2019). Figure 5 provides an illustration of how the various ingredients of the model interact with one another.

5.1 Households

The representative household maximizes lifetime utility:

$$\max_{\{C_t, N_t, B_{t+1}, D_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{(C_t - \varphi H_t)^{1-\sigma}}{1-\sigma} - \theta \frac{N_t^{1+\eta}}{1+\eta} + \phi \ln \left(\frac{D_t + B_t^T}{P_t} \right) \right) \tag{1}$$

subject to the budget constraint:

$$P_t C_t + B_{t+1} + D_t + B_t^T \leq W_t N_t + \Psi_t + \Phi_t + (1 + i_{t-1}^B) B_t + (1 + i_{t-1}^D)(D_{t-1} + B_{t-1}^T) - P_t T_t$$

Where E_0 is the expectation operator at time 0, β is the discount factor and $0 < \beta < 1$. The household gets utility from consumption C at price P, and disutility from labor N, which pays a wage of W. The household gets liquidity services from deposits D and treasury notes B^T. The firm also has access to the privately issued risk-free bond B, which is illiquid, unlike deposits and treasury notes. T is the lump sum tax paid to the government.

The household owns the final goods firm, from which it receives profits Ψ_t. The household also own the bank, which yields the household profits Φ_t.

5.2 Intermediate Goods Producers

In the model, there are both intermediate and final goods producers. The setup for the final goods producer is standard. However, the intermediate good firm has a particular aspect about it that allows us to connect it to the banking sector.

Each intermediate goods producer $j \in [0, 1]$ operates a constant returns to scale production function:

$$Y_t(j) = A_t N_t(j),$$

where Y_t and A_t are output, and an aggregate productivity shock, respectively.

The firm's profit function is given by:

$$\Psi_t(j) = P_t(j)Y_t(j) - (1 + i_t^L) [1 - \phi_{d,t+1}(j)] W_t N_t(j).$$

This is where we connect the banking sector and the production side of the economy. The firms finance their inputs via loans from the bank:

$$W_t N_t(j) = L_t(j)$$

The idea is that firms finance their inputs via loans that they must repay with interest to the bank. However, there is a risk that the firms will not repay the loans that is captured in the default variable $\phi_{d,t+1}$ that depends on both loans and the interest rate on loans.

So, we get:

$$\Psi_t(j) = P_t(j)Y_t(j) - (1 + i_t^L) [1 - \phi_{d,t+1}(j)] L_t(j).$$

which allows us to endogenize default:

$$\phi_{d,t+1} = \max\left(1 - \frac{Y_t}{(1+i_t^L)\ell_t}, 0\right) \tag{2}$$

This shows that the default rate is bounded between 0 and 1, and is increasing in loans ℓ_t and the interest rate on loans i_t^L. The default rate is also decreasing in output.

5.3 Bank Sector

The banking sector setup comes from Benigno and Benigno (2021). I add an adjustment cost for loans that captures the inertia in bank lending for reasons discussed in section 4.1. Then I derive equations for bank balance sheet variables like loans, deposits, reserves, etc. The Beningo et al. paper does not derive or look at IRFs for these variables in response to policy shocks. I show how to derive these equations in the appendix.

The bank chooses $\{L_t, B_t, R_t, D_t, X_t\}$ to maximize:

$$\Phi_t = E_t\Big\{\Lambda_{t+1}\Big[(1+i_t^L)(1-\phi_d)L_t + (1+i_t^B)B_t + (1+i_t^R)R_t - (1+i_t^D)D_t\Big]\Big\} - X_t - \frac{\phi_x}{2}(L_t - L_{t-1})^2 \tag{3}$$

subject to:

$$L_t + B_t + R_t = D_t + (1 - f(\delta_t))X_t, \quad \text{where} \quad \delta_t = \frac{L_t}{X_t}, \tag{4}$$

$$f(\delta_t) = \frac{\alpha}{2}\delta_t^2, \tag{5}$$

$$R_t \geq \rho D_t, \quad 0 \leq \rho < 1. \tag{6}$$

Λ_{t+1} is a stochastic discount factor equal to $\beta\left(\frac{C_{t+1}}{C_t}\right)^{-\sigma}\frac{P_t}{P_{t+1}}$. The bank supplies loans L and charges interest rate i^L for said loans. It also holds privately issued bonds B, earning interest rate i^B. The bank also keeps reserve balances R with the central bank that are remunerated at the policy rate i^R. The household holds deposits D with the bank, which the bank remunerates the household at a rate of i^D.

The bank is also able to raise equity X, subject to a cost $f(\delta)$, where δ denotes bank leverage, and the cost of raising equity is increasing in leverage. The bank is subject to a reserve requirement: $R > \rho D$, meaning its reserves must exceed some specified fraction of its deposits. The banks profit function incorporates an adjustment cost for loans via the final term: $\frac{\phi}{2}(L_t - L_{t-1})^2$.

Finally, the bank faces a capital requirement. There is a regulatory ceiling such that:

$$\frac{L_t}{X_t} \leq \bar{\delta} \tag{7}$$

Therefore, if the bank enters a state where $\delta_t > \bar{\delta}$, the capital constraint will bind and force the bank to de-lever. This setup delivers a one-sided occasionally binding constraint à la Guerrieri and Iacoviello (2015), which is what banks face in reality: bank regulations such as those under Basel III set capitalization constraints (the inverse of leverage) from below, but not from above. This does most of the work at the early horizons in generating the asymmetry in credit issuance in the model.

As we will see below, the monetary policy shock will enter the bank's balance sheet via its effects on the interest rate the central bank offers on reserves balances. The effects of the shock to the interest rate on reserves will ripple through the banking system, affecting all other interest rates to varying degrees.

5.4 Forward Guidance Shock

The central bank follows a standard Taylor Rule, adjusting the nominal interest rate in response to inflation and output deviations:

$$1 + i_t^R = (1 + i^R)^\rho \left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi} \left(\frac{Y_t}{Y^*}\right)^{\phi_Y} e^{\phi_t}. \tag{8}$$

Where i^R, Π^*, and Y^* are the steady state values of the central bank's policy rate, inflation, and output, respectively. Shocks to the Fed's policy rate enter in through ϕ_t.

The central bank communicates policy deviation new up to 20 periods before the policy shock manifests itself in reality. Deviations in the policy rate can be characterized by news shocks according to:

$$\phi_t = \sum_{i=0}^{20} \hat{\psi}_{r,t-i}^i \tag{9}$$

Meaning that forward guidance can extend 20 quarters ahead, which is consistent with the time frame I use to calculate my IRFs from the data.

w_t is a weight that decays geometrically over time, meaning it assigns ever smaller weights to more distant quarters. Therefore, agents will place less weight on distant forward guidance shocks when it comes to their current decision making. w_t is specified as: $w_t = (1-\rho_w)\rho_w^t$. The calibration for ρ_w is provided in the next section.

Let $\hat{\psi}_{r,t}$ denote one's belief (or prior) about the true value of $\psi_{r,t}$, which represents the true policy innovation at time t. $\hat{\psi}_{r,t}$ is defined as follows:

$$\hat{\psi}^j_{r,t} = \hat{\psi}^j_{r,t-1} + \kappa_j(s^j_t - \hat{\psi}^j_{r,t-1}), \quad j = 1, ..., 20. \tag{10}$$

Here, κ_j represents the Kalman gain with respect to horizon j, and $s^j_t - \hat{\psi}^j_{r,t-1}$ is the forecast error.

The forward guidance shock ψ^{FG}_t is defined as a set of signals:

$$\psi^{FG}_t = [s^0_t \;\; s^1_t \dots s^{20}_t]$$

The current policy rate is perfectly observed:

$$s^0_t = \psi^0_{r,t}$$

While future deviations are unobserved, i.e. for $j \geq 1$

$$s^j_t = \psi^j_{r,t} + \nu_t, \qquad \nu_t \sim N(0, \sigma^2_\nu) \tag{11}$$

Hence, the Kalman gain $\kappa = \frac{\sigma^2_\psi}{\sigma^2_\psi + \sigma^2_\nu}$, where σ^2_ψ is the variance of the agent's prior. Importantly, the agent does not know σ^2_ν, rather, they believe it to be somewhere in the range of $\sigma^2_\nu \in [\underline{\sigma}^2, \overline{\sigma}^2]$. The value of σ^2_ν that the agent ends up entertaining depends upon the realized value of s_t.

5.5 Government

One final set of ingredients involved in the model is a government with the following budget constraint:

$$B_t^T = (1 + i_t^D) B_{t-1}^T - T_t \tag{12}$$

Which denotes the government's primary surplus. Further, assume that the government sets the tax policy in the following manner:

$$T_t = \tau Y_t, \qquad 0 < \tau < 1 \tag{13}$$

So that total lump sum taxes are proportional to output.

The remaining ingredients are standard, with a final goods firm that combines a continuum of differentiated intermediate goods according to a Dixit-Stiglitz aggregator. As shown above, each intermediate good producer uses labor to produce their output. The wages paid to laborers are financed via loans from the bank, plus interest. There is an endogenously determined probability of default that is dependent upon the existing level of output, loans, and interest owed on loans. Prices are also made to be sticky via the classic Calvo pricing setup.

5.6 Model Calibration

I use Bayesian methods à la Smets and Wouters (2007) to estimate the model. I estimate several parameters following adjacent strands of literature with relevant citations. Then I present the priors and their posterior estimates. I use 16 observables from FRED to estimate the model, which includes: real GDP, real consumption, consumer price inflation, median usual weekly real earnings, nonfarm employment, bank equity capital, reserve balances with Federal Reserve banks, 3-month rates and yields on certificates of deposit, tier 1 leverage capital, total loans and leases for commercial banks, commercial bank holdings of treasury and agency securities, deposits for commercial banks, bank-prime loan rate, nonfarm labor productivity, the federal funds rate, and the net charge-off rate on total loans and leases.

Table 1: Calibrated Parameters

Parameter	Value	Description	Source
β	0.99	Discount Factor	–
σ	2	Risk Aversion	–
η	2	Frisch Elasticity of Labor Supply	–
θ	1	Labor Disutility Weight	–
ϕ	0.1	Liquidity Services Weight	–
ρ	0.1	Reserve Ratio	Benigno & Benigno (2021)
κ	0.5	Kalman Gain	Coibion & Gorodnichenko (2015)
τ	0.3	Tax Rate	OECD Centre for Tax Policy & Administration

Table 6 in the appendix presents the parameter estimates using the Metropolis algorithm, running four chains, each with 250,000 draws. The calibration for the parameters listed in Table 1 above are standard. However, beginning with ϕ_x, we encounter parameters that are more difficult to calibrate. The parameter ϕ_X is an adjustment cost parameter for loans. Although the specific adjustment cost for loans, $\frac{\phi_x}{2}(L_t - L_{t-1})^2$, is unique to this paper, I inform my choice of ϕ_x through other papers that incorporate financial market frictions, such as Christiano et al. (2014).

We know that the cost of raising equity, $f(\delta_t)$ in the model, is increasing in leverage. Since I specify this function as $\frac{\alpha}{2}\delta_t^2$, knowing how much each increase in leverage affects equity costs will inform us as to the calibration of α. Corbae and D'Erasmo (2019) calibrate a banking model where large banks face a marginal cost of equity issuance of about 2.5% of the funds raised. A 2.5–5% issuance cost means that for each \$1 of new equity capital raised, the bank forfeits \$0.025–\$0.05 in fees, higher capital costs, and dilution costs. In terms of what this means for our calibration, if we assume an equity issuance cost of roughly 2.5%, then using $f(\delta) = \frac{\alpha}{2}\delta^2$ and normalizing $\delta = 1$, we can set $\frac{\alpha}{2} \approx 0.025$, which yields $\alpha \approx 0.05$.

The reserve ratio is set at 10%. This is consistent with what the reserve ratio in the

U.S. banking system has been historically.[8] Benigno and Benigno (2021) experiment with varying levels of reserve requirements, with 10% being the lowest level they model with.

The parameters associated with the Taylor rule are fairly standard, but are taken directly from Campbell et al. (2019). The last parameter source is the Kalman Gain κ, which is calibrated to 0.5. This is consistent with similar calibrations in the literature, as in Coibion and Gorodnichenko (2015) who compute $\kappa = 0.46$.

Below, I explain that I use occasionally binding constraints à la Guerrieri and Iacoviello (2015) to generate the asymmetric pattern. In order to calibrate the model, I first estimate the linear, standalone model for the parameters, before implementing the occasionally binding simulations.

6 Model Results

6.1 Mechanism

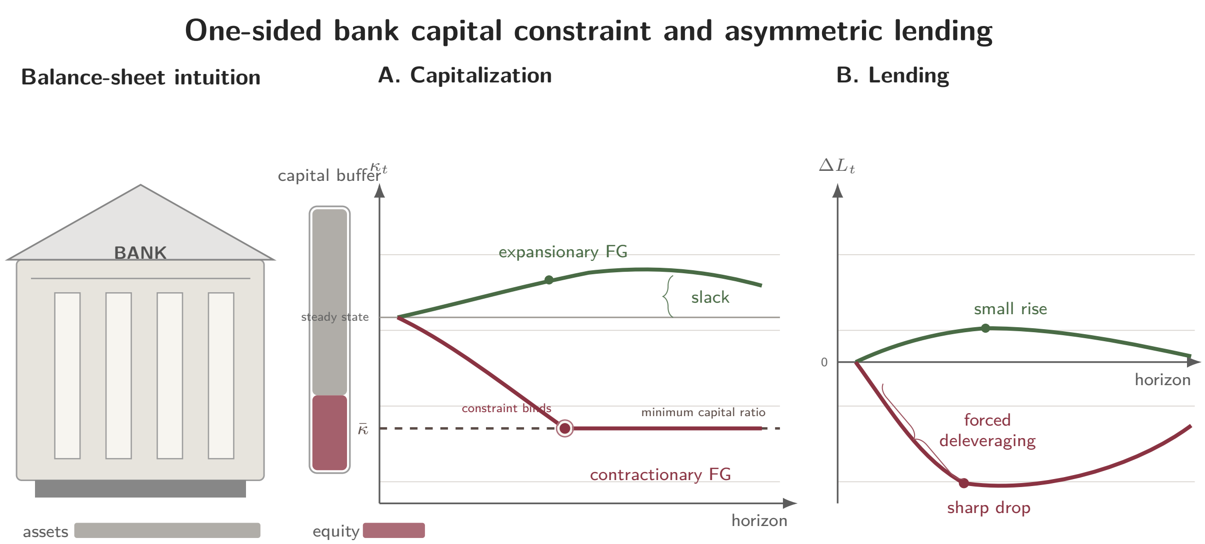


Figure 6: Model Mechanism

[8]From about 1993 until March 2020, the required reserve ratio, set by the Fed, was 10%. Since the onset of the pandemic, the reserve ratio has been dropped to zero.

Figure 6 represents an illustration of the core mechanism behind the results of the model. A bank faces a capital constraint determined by the level of equity relative to their assets. This constraint is binding in only one direction, which is from above (below) in the case of leverage (capitalization). Expansionary forward guidance creates an improvement in capitalization, while contractionary forward guidance harms it. Therefore, banks only approach the binding constraint in the contractionary environment, which explains the forced delevering and consequent asymmetric response of lending to the contractionary shock versus expansionary shock.

6.2 Capital Constraints

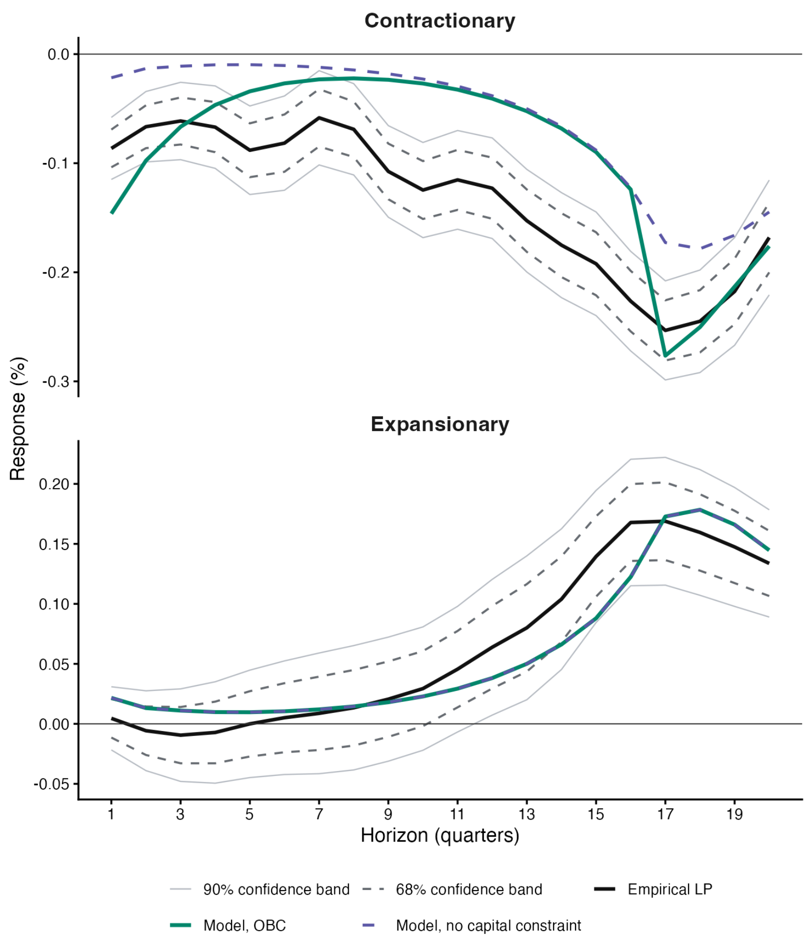


Figure 7: Model and Data: Responses to Forward Guidance Shocks

The model reproduces the asymmetry via the binding capital constraint, as shown in Figure 7.

The data is represented by the solid black lines, with corresponding 68% and 90% confidence bands. This is the same data as presented in Figure 2. The purple dashed line represents how the model responds to contractionary (top panel) and expansionary (bottom panel) forward guidance shocks without capital constraints. The solid lines represent the situation where the bank faces the capital constraints and they can possibly bind. When the solid line deviates from the dashed line, that represents the point where the constraint binds, which, in this case happens in the contractionary direction only. This forces a sharp decline in lending as the bank is required to de-lever immediately. This process of de-levering results in a contraction of credit that, in absolute magnitude, exceeds the expansion of credit after an expansionary FG shock.

Additionally, the endogenous probability of default by firms who have borrowed from banks rises sharply, adding to the immediate downward pressure on lending in the near horizon. The adjustment costs of the bank generate inertia that induce it to adjust its lending only gradually over medium-term horizons (as observed in the data and explained in section 4.1), until quarter 15 when broader deterioration in economic conditions pushes down bank equity to a point where, once again, capital constraints bind and force another sharp contraction in credit.

In the expansionary case, leverage falls on impact, so capital constraints are not relevant to the bank's initial credit issuance. Eventually leverage begins to rise as banks expand credit starting at horizon 11, but they never hit capital constraints in this expansion, which means that constraint never binds. These results may provide additional perspective on the question of countercyclical capital buffers (CCyB) introduced in Basel III. Figure 7 serves not only to reproduce the dynamics of the data, but also to simulate what would happen if banks were well-capitalized when facing a contractionary shock; the dashed line serves to show that the asymmetry disappears once capital constraints no longer bind.

Figure 8 shows how lending responds to a given contractionary FG shock by the initial level of bank capitalization. Holding assets fixed, I test how lending responds to the shock as equity deteriorates along a spectrum of equity levels. The color gradient begins with yellow representing the best capitalized banks and ends with purple representing the worst capitalized banks.

Consistent with the mechanism proposed, lending falls more sharply in the wake of a given contractionary forward guidance shock as bank capitalization deteriorates.

The purpose of the foregoing section was to illustrate the mechanism behind the asymmetry in credit's response to expansionary versus contractionary forward guidance shocks. Reiterating it plainly, forward guidance contracts credit more sharply than it expands it due to capital constraints binding in one direction — the contractionary direction. The constraint binds because leverage rises sharply after a contractionary FG shock. The rise in leverage is owing to equity dropping faster than assets. As leverage dynamics bring the bank to the point of the capital constraint, the constraint binds, which forces banks to delever by contracting credit strongly. These same constraints are never faced by banks in the expansionary direction; hence the asymmetry. Now I turn to an analysis of what forward guidance does to credit in a ZLB episode.

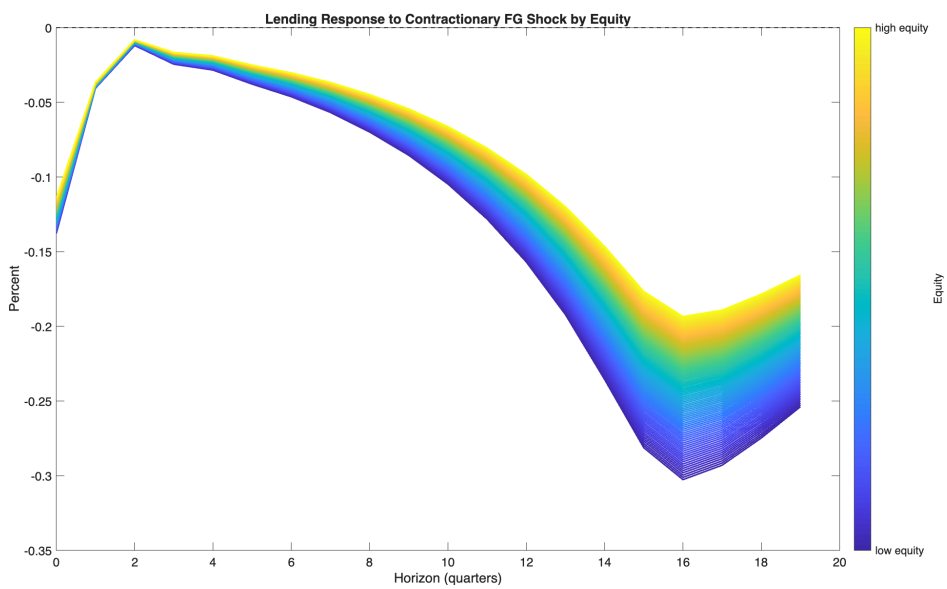


Figure 8: Lending by Equity Level

Finally, holding the absolute magnitudes of the expansionary and contractionary shocks constant, I shrink the distance betwen $\bar{\delta}$ and the bank's initial value of leverage δ before a shock hits. As the distance between the binding capital constraint and steady state leverage falls, bank balance sheets have less of a buffer to absorb shocks. I examine five gaps of varying magnitude.

6.3 ZLB Periods

Of particular interest when examining the effects of forward guidance is how forward guidance can affect economic activity at the zero-lower-bound (ZLB). We are especially interested in the potential of forward guidance to stimulate the economy when adjustments to the policy rate are no longer and option. To this end, I examine the efficacy of a forward guidance expansion at the ZLB, once again, using Guerrieri and Iacoviello (2015) to simulate the binding policy rate.

An expansionary forward guidance shock at the ZLB is effectively communication from the central bank that its policy rate will be kept at zero for longer, or that the policy rate hike will take place later than previously expected. To this end, I simulate two forward guidance *contractions*: one that is communicated to take place one quarter in the future, and a second that is communicated to take place four quarters into the future. The difference between the effects of the second and first announcements will be taken to represent our expansionary forward guidance at the ZLB.

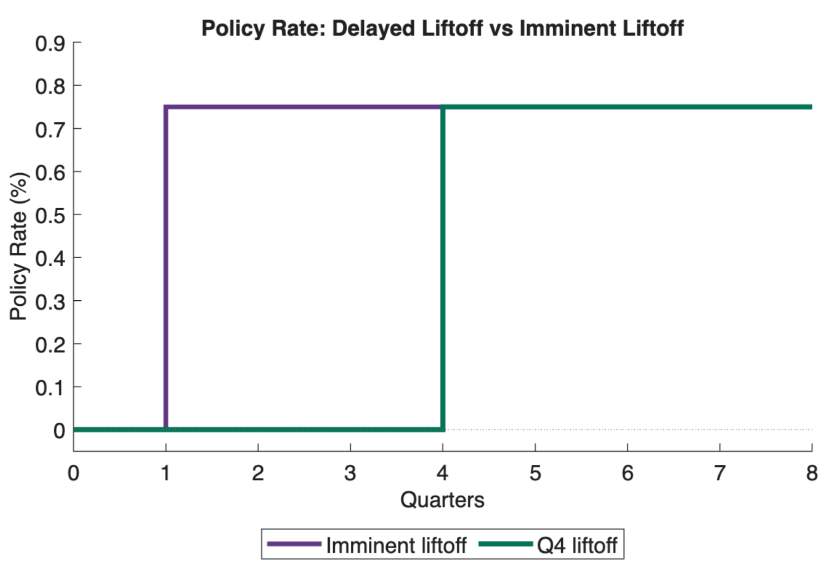


Figure 9: Policy Rate in Response to a FG Shock at ZLB

The policy rate in the ZLB experiment, which is initially expected to rise imminently, but is then announced to rise three quarters later than expected with a permanent increase of 75 basis points.

Specifically, we have:

$$\Delta Y_{4,1} = Y_{t,\varepsilon_t^4} - Y_{t,\varepsilon_t^1} \equiv Y^{ZLB}_{t,\varepsilon_t^{FG}}$$

Where Y_{t,ε_t^4} is the impulse response at time t to an four-period-ahead contractionary FG shock ε_t^4.

Figure 9 illustrates the change in policy rates. Initially, the policy rate i^R is expected to be held at zero until quarter 1, at which point it will be raised 75 basis points (purple line). The central bank then announces that instead, the policy rate will be raised 75 basis points in quarter 4 (green line).

Below, I plot what happens to lending in response to this expansionary forward guidance shock at the ZLB. In Figure 10, lending is stimulated. However, consistent with McKay et al. (2016), the magnitude of the response is quite modest. There is a rise in lending that peaks in quarter three at about 4.5 basis points, which quickly tapers off. The small magnitudes of these responses are fairly informative about the power of forward guidance at the zero bound — expansionary forward guidance *does* stimulate lending activity at the ZLB, but only weakly. This is owing to the fact that the primary driver of economic activity in this scenario is the financial sector.

More specifically, an important transmission channel is the increase in bank equity, corresponding to a drop in bank leverage, which is coupled with a temporary credit expansion. However, the jump in credit is very modest and, therefore, so are the effects on output. This highlights that not only are bank balance sheets a major economic driver at the zero bound, but that they constrain the effects of expansionary forward guidance at the ZLB, which suggests that forward guidance alone is unlikely to generate large effects in a liquidity trap.

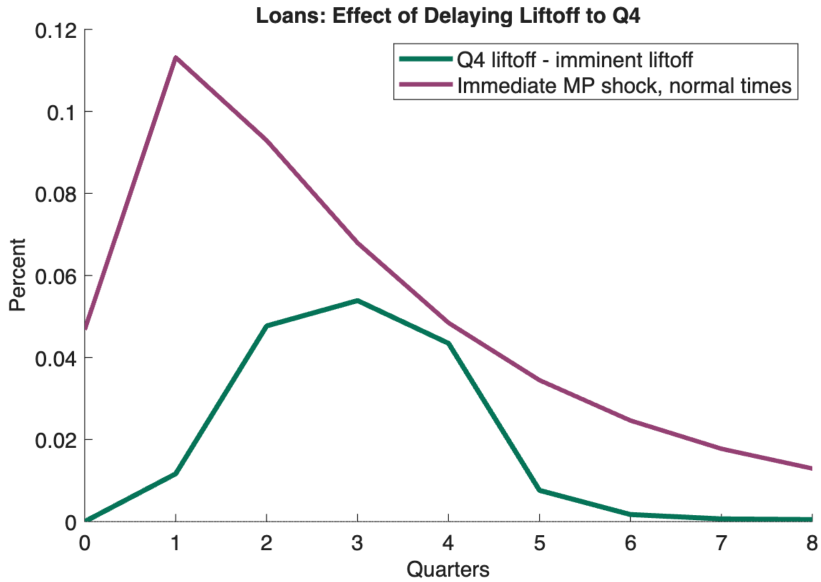


Figure 10: Expansionary FG Shock at ZLB

These are the impulse responses of loans and output in going from a an environment where agents expect credit tightening a year into the future to two years into the future.

Given the importance of bank equity in the transmission of forward guidance to lending, I test the efficacy of an exogenous injection of equity to the banking sector to examine whether this can provide a level of stimulus that, when paired with expansionary forward guidance at the ZLB, would generate an expansion similar in magnitude to that of an immediate monetary policy shock.

This scenario is identical to that which produced Figure 10, with the difference that the central bank (or government) provides balance-sheet relief to the banks by injecting equity. The real-world analogy would be the Troubled Assets Relief Program (TARP) in the wake of the 2008 financial crisis, which initially apportioned $700 billion to the Treasury to purchase rapidly depreciating assets (in addition to providing relief to the auto industry.[9]

The outcome of this experiment is illustrated in Figure 11. Our reference point is the black line that represents the same FG expansion (with zero equity injections: "Equity = +0.00%) as Figure 10. Here, the exercise consists in injecting equity to the banking system at the same time that the forward guidance expansion is announced. The magnitudes of

[9]TARP's initial apportionment was $700 billion, much of which was clawed back by Dodd-Frank, for a total authorized limit of $475 billion in relief. About $245 billion was used for bank stabilization, which is the relevant operation for our purposes here.

the injections range from 0% of steady-state equity (baseline) to a maximum of 20% of steady-state equity.

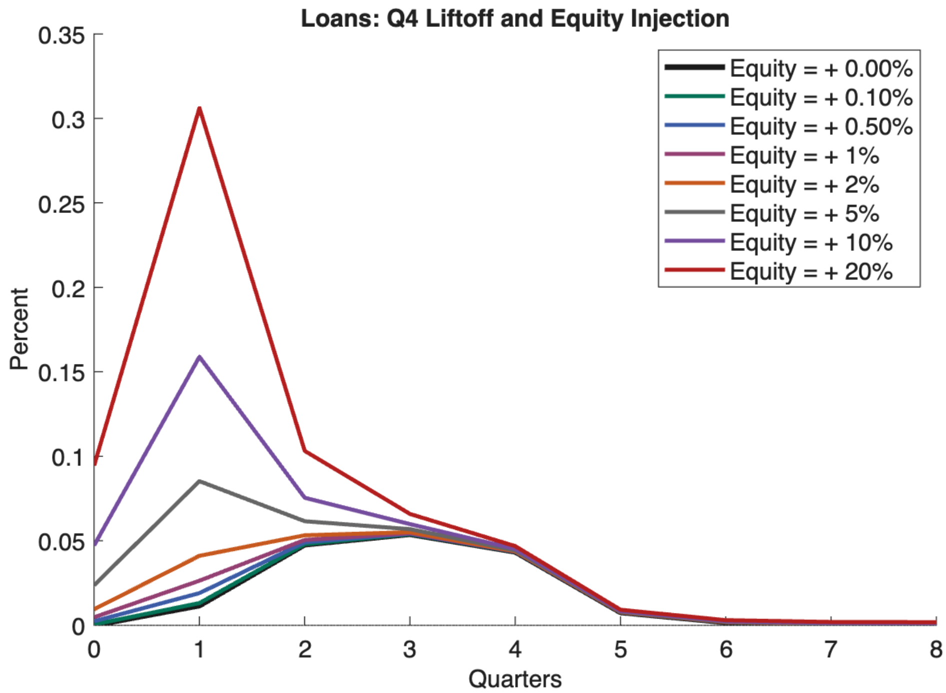


Figure 11: Expansionary FG Shock with Simultaneous Equity Injections at ZLB

The interpretation is that expansionary forward guidance begins to have an expansionary effect on lending that resembles the magnitudes of comparable immediate-term monetary policy target shocks once the equity injection approaches about 10% of steady-state bank equity. For reference, according the Federal Reserve's H.8 release on "Total Equity Capital" of FDIC-insured banks, on average in the year 2025, is estimated to be about $2.55 trillion. This means that a 10% equity injection would approximately amount to $255 billion, which is roughly the magnitude of the bank-stabilization package implemented in TARP.

Therefore, under crisis scenarios, expansionary FG (supplemented with equity injections) at the ZLB can generate similar expansionary effects for credit as a contemporaneous monetary policy shock; but barring an equity injection of historic magnitude, equity injections have little efficacy in stimulating credit. Even in the case of a direct equity injection of reasonable magnitude, expansionary forward guidance is still limited in its ability to stimulate credit, and especially so at the zero lower bound, where it is most relevant.

7 Conclusion

Forward guidance has become an increasingly important tool of central banks around the world. In countries with credible central banking systems, forward guidance can be a particularly effective tool in stimulating the economy. This paper has analyzed the effect that forward guidance has on commercial bank balance sheets. I've analyzed the effects of both expansionary and contractionary forward guidance shocks.

I find that banks' balance sheets are much slower to respond to expansionary forward guidance than they are to contractionary forward guidance, even when the shocks are of the same magnitude (absolute value). Second, the magnitude of the adjustments themselves are smaller during an expansion than during a contraction, which indicates that expansionary forward guidance is less effective in stimulating bank activity than contractionary forward guidance is in tightening bank balance sheets. I provide evidence that this is owing ultimately to capital requirements. Contractionary forward guidance shocks cause asset prices, and therefore, bank equity to fall in an absolute magnitude that is greater than their rise following an expansionary forward guidance surprise. When asset prices fall, banks also become less well-capitalized. As they inch closer to minimum capital requirements, they're forced to substantially reduce credit. In the wake of expansionary shocks, these requirements do not force behaviors on the part of banks in the way that they do in a contraction. Therefore we observe an asymmetry in credit.

One of the principal interests that people have in forward guidance policy is its ability to *stimulate* the economy, especially in the case of a liquidity trap. My findings show that although the central bank appears capable of stimulating credit expansions and output during the ZLB period, the responsiveness of banks to this kind of stimulus is fairly muted, much more so than what we observe in response to contractionary forward guidance shocks. Therefore, forward guidance is quite effective at contracting credit, but weak at stimulating it, which is precisely when forward guidance is supposed to be most useful.

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Cross-Sectional Analysis

Table 2: Tier 1 Capital Tightness and Bank Lending Growth

	(1) Baseline	(2) + AOCI	(3) + AOCI & Equity
Dependent variable:		$\Delta \log(\text{Loans}_{i,t})$	
Tier 1 Ratio$_{i,t-1}$	0.210**	0.210**	0.184*
	(0.075)	(0.075)	(0.082)
log(Assets$_{i,t-1}$)	-0.045***	-0.044***	-0.029***
	(0.009)	(0.009)	(0.005)
Deposits/Assets$_{i,t-1}$	-0.040	-0.048	0.047
	(0.065)	(0.067)	(0.027)
RWA/Assets$_{i,t-1}$	-0.007	-0.007	-0.027
	(0.034)	(0.034)	(0.040)
Nonaccrual Loans/Loans$_{i,t-1}$	-0.307**	-0.300**	-0.304**
	(0.103)	(0.104)	(0.105)
Past Due Loans/Loans$_{i,t-1}$	-0.296*	-0.290*	-0.300*
	(0.139)	(0.138)	(0.130)
AOCI/Assets$_{i,t-1}$		-0.165**	-0.628*
		(0.051)	(0.260)
Equity/Assets$_{i,t-1}$			0.518*
			(0.261)
Bank fixed effects	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes
Observations	118,496	117,558	117,143
R^2	0.243	0.243	0.249
Within R^2	0.109	0.109	0.116

Notes: The dependent variable is quarterly loan growth, measured as $\Delta \log(\text{Loans}_{i,t})$. All specifications include bank fixed effects and quarter fixed effects. Standard errors are clustered at the bank level and are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 3: Bank Capitalization and Loan Growth

	Linear baseline	Piecewise threshold	Low-capital interaction	Near-threshold interaction
Lagged Tier 1 ratio	0.210**		0.210**	0.210**
	(0.075)		(0.075)	(0.075)
Distance below threshold, $d^-_{i,t-1}$		−1.888***		
		(0.493)		
Distance above threshold, $d^+_{i,t-1}$		0.209**		
		(0.075)		
Low-capital indicator			−0.047**	
			(0.017)	
Lagged Tier 1 ratio × Low capital			0.434**	
			(0.153)	
Near-threshold indicator				0.031
				(0.051)
Lagged Tier 1 ratio × Near-threshold				−0.434
				(0.622)
Bank fixed effects	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	117,558	117,558	117,558	117,558
Within R^2	0.109	0.109	0.109	0.109

Notes: The dependent variable is quarterly loan growth, $\Delta \log(Loans_{i,t})$. Standard errors, clustered at the bank level, are reported in parentheses. The piecewise specification defines the distance from the Tier 1 capital threshold as $d_{i,t-1} = Tier1Ratio_{i,t-1} - \tau$, with $d^-_{i,t-1} = \max\{-d_{i,t-1}, 0\}$ and $d^+_{i,t-1} = \max\{d_{i,t-1}, 0\}$. Thus, $d^-_{i,t-1}$ measures the size of the capital shortfall below the threshold, while $d^+_{i,t-1}$ measures the size of the capital buffer above the threshold. The low-capital indicator equals one for banks in the bottom quartile of the lagged Tier 1 capital-ratio distribution. The near-threshold indicator equals one for banks within one percentage point of the Tier 1 threshold. Controls include lagged log assets, deposits-to-assets, risk-weighted-assets-to-assets, nonaccrual loans-to-loans, past-due loans-to-loans, and AOCI-to-assets. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 4: Forward Guidance Shocks and Bank Lending: Heterogeneity by Tier 1 Capital Buffer Bins

	(1) Contractionary FG	(2) Expansionary FG
Dependent variable:	$\Delta \log(\text{Loans}_{i,t})$	
FG $\times$ Below requirement	-0.007^{**}	-0.001
	(0.002)	(0.001)
FG $\times$ Buffer bin $[0, 1)$ pp	-0.003	-0.002
	(0.002)	(0.001)
FG $\times$ Buffer bin $[1, 2)$ pp	-0.001	-0.001^{*}
	(0.001)	(0.000)
FG $\times$ Buffer bin $[2, 4)$ pp	-0.000	-0.000
	(0.000)	(0.000)
Controls	Yes	Yes
Bank fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Observations	117,558	117,558
R^2	0.175	0.174
Within R^2	0.028	0.028

Notes: This table reports heterogeneity in the lending response to forward-guidance (FG) surprises by banks' Tier 1 *capital buffer*, measured relative to the regulatory requirement. The dependent variable is quarterly loan growth, $\Delta \log(\text{Loans}_{i,t})$. The contractionary shock is defined as the positive part of the quarterly FG surprise (FG_t^{contr}), and the expansionary shock is defined as the positive magnitude of the negative FG surprise (FG_t^{exp}). Each coefficient corresponds to an interaction between the shock and an indicator for the bank's Tier 1 capital-buffer bin at $t-1$, with the omitted category being banks with at least 4 percentage points of Tier 1 capital buffer (i.e., Tier 1 ratio $\geq \tau + 4$pp). All specifications include bank and quarter fixed effects and the baseline lagged bank controls (log assets, deposits/assets, RWA/assets, nonaccrual loans/loans, past due loans/loans, and AOCI/assets). Standard errors clustered at the bank level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

A Appendix

A.1 Empirical Design

Federal funds rate futures are a financial instrument whose payout is calculated by comparing the contract price to the average federal funds effective rate over the month preceding expiration of the contract. The price of the contract is calculated, simply, as 100 - Expected Federal Funds Rate. If the expected federal funds rate at the end of the month is lower than anticipated when the contract was written, then the holder profits (the holder loses money if rates end up higher than expected at origination of the contract).

The surprise captured by the change in the federal funds rate target in the aftermath of an FOMC announcement can be specified as[10]:

$$mp1_t = (ff1_t - ff1_{t-\Delta t})\frac{T_1}{T_1 - \tau_1}$$

where T_1 is number of days in the expiration month, and τ_1 is the day of the FOMC announcement in month T_1. $ff1_t$ can simply be calculated as $ff1_t = 100 - P_{ff,t}$ Where $P_{ff,t}$ is the price at time t of federal funds futures contract. $ff1_{t-\Delta t}$ is specified as:

$$ff1_{t-\Delta t} = \frac{\tau_1}{T_1}p_0 + \frac{T_1 - \tau_1}{T_1}E_{t-\Delta t}p_1 + \xi_{t-\Delta t}$$

Here, p_0 denotes the the federal funds that that has prevailed up to the current point of month, and p_1 is the expected rate for the remainder of the month. $\xi_{t-\Delta t}$ is a risk premium.

A similar set of procedures allows one to identify the revision in expectations about what the federal funds rate target will be following the next FOMC announcement[11]:

$$mp2_t = \left[(ff2_t - ff2_{t-\Delta t}) - \frac{\tau_2}{T_2}mp1_t\right]\frac{T_2}{T_2 - \tau_2}$$

[10]All of the following mathematical details can be found in Gürkaynak et al. (2005) and are derived from there.

[11]The remainder of the details about the construction of Eurodollar and Treasury Futures can be found in GSS (2005)

In identifying the factors one uses the following factor specification:

$$X = F\Lambda + e$$

Where Λ is a 2 x n matrix of factor loadings, F is a 204 x 2 matrix of unobserved factors. The e term denotes white noise.

In performing the rotation of the two factors, I define the 204 x 2 matrix Γ:

$$\Gamma = F\Upsilon$$

where

$$\Upsilon = \begin{bmatrix} \rho_1 & \lambda_1 \\ \rho_2 & \lambda_2 \end{bmatrix}$$

Here, columns are normalized to unit length. Then one restricts the two factors Γ_1 and Γ_2 to be orthogonal:

$$E(\Gamma_1\Gamma_2) = \rho_1\lambda_1 + \rho_1\lambda_1$$

After this, it is ensured that Γ_2 has no influence on the surprise to the current federal funds rate in the following way: denote α_1 and α_2 as the loadings of the current federal funds rate surprise on F_1 and F_2, respectively. Then, since

$$F_1 = \frac{1}{\rho_1\lambda_2 - \rho_2\lambda_1}[\lambda_2\Gamma_1 - \rho_2\Gamma_2]$$
$$F_2 = \frac{1}{\rho_1\lambda_2 - \rho_2\lambda_1}[\rho_1\Gamma_2 - \lambda_1\Gamma_1]$$

which yields:

$$\alpha_2\rho_1 - \alpha_1\rho_2 = 0$$

At which point, one can identify Υ.

A.2 Buffer Bin LP Sample Details

Table 5: Estimation support by Tier-1 capital buffer bin

Tier-1 buffer bin	Bank-quarters	Distinct banks	Share of sample
Below requirement	2,640	663	0.51%
$[0, 1)$ pp	1,279	532	0.25%
$[1, 2)$ pp	2,566	870	0.49%
$[2, 4)$ pp	44,048	4,216	8.44%
$\geq$ 4 pp (reference)	471,373	9,770	90.32%
Total	521,906	—	100.00%

Notes: Buffer bins are defined by each bank's Tier-1 capital ratio relative to the 6% regulatory requirement (buffer = Tier1 − 6pp), lagged one quarter. The $\geq$ 4pp bin is the omitted reference group in the interaction local projections. *Bank-quarters* and *distinct banks* count the observations and unique banks falling in each bin over the estimation sample; distinct-bank counts do not sum across bins, since banks move between bins over time (nearly all banks appear in the reference bin at some point). *Share of sample* is the fraction of total bank-quarter observations; shares may not sum to 100 due to rounding.

A.3 Additional IRFs

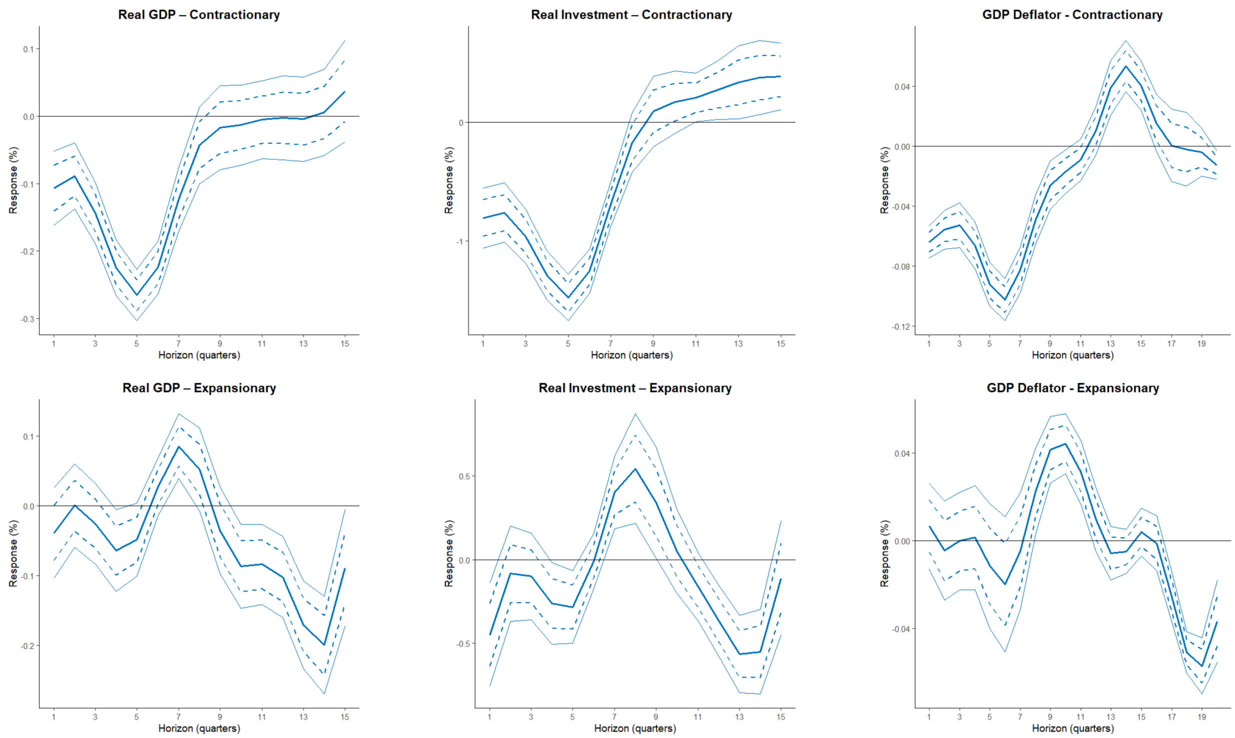
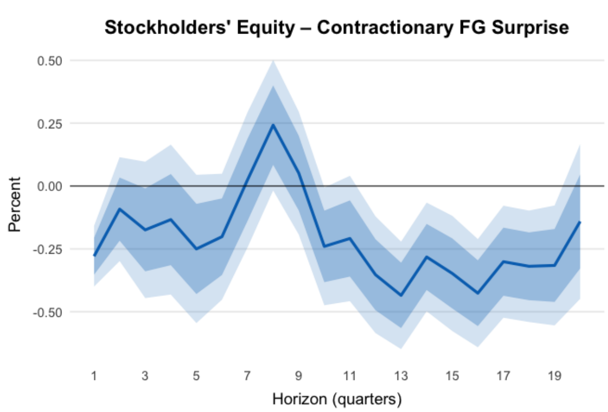
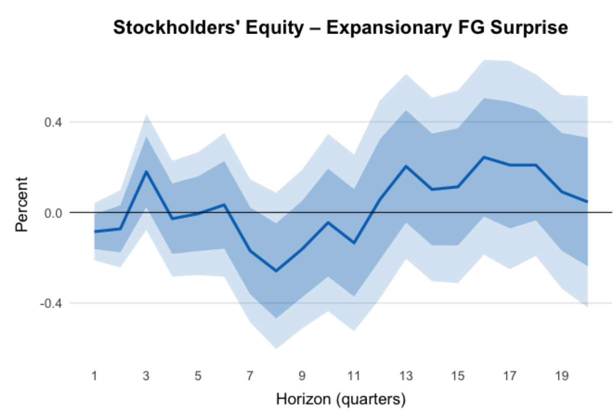


Figure 12: IRFs of Aggregates

The figures are the impulse responses of various aggregate variables to the one standard deviation contractionary forward guidance shock. The inner dashed band corresponds to the 68% confidence band, while the. outer solid band corresponds to the 90% confidence band.



(a) Contractionary FG surprise



(b) Expansionary FG surprise

Figure 13: Impulse responses of stockholders' equity to forward-guidance surprises

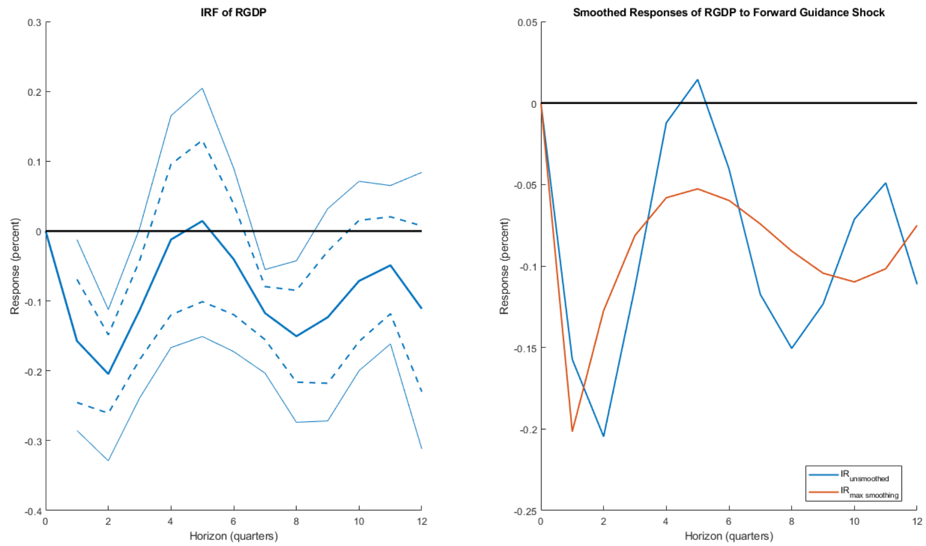


Figure 14: Real GDP Response to a Contractionary Forward Guidance Shock

The figures respresent the impulse response of real GDP to a one standard deviation contractionary forward guidance shock. For the left hand figure, the inner dashed band corresponds to the 68% confidence band, while the outer solid band corresponds to the 90% confidence band. The right hand figure shows how the impulse response dynamics change as we go from a local projections IRF (IR_{lp}), to a smoothed local projections plot (IR_{slp}), and maximally smoothed response ($IR_{slp,maxpen}$).

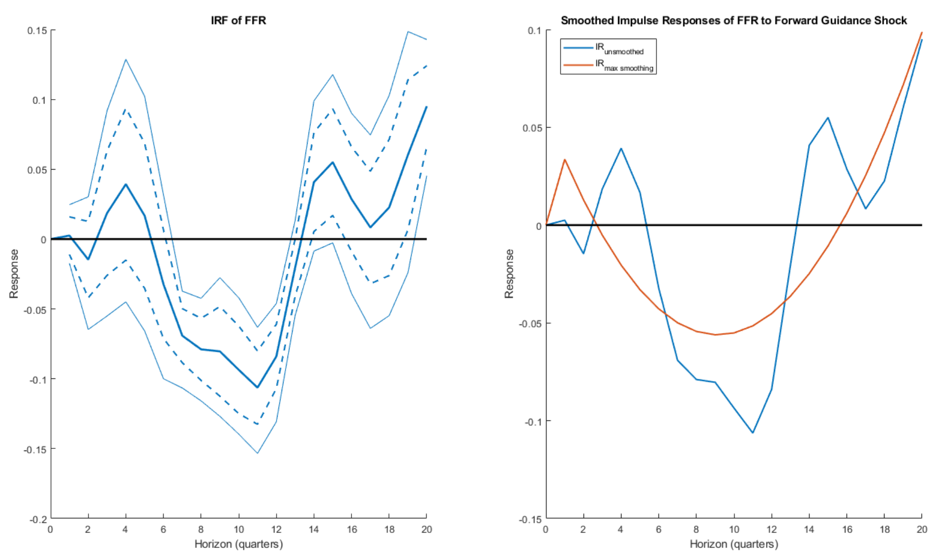


Figure 15: Federal Funds Rate Response to a Contractionary Forward Guidance Shock

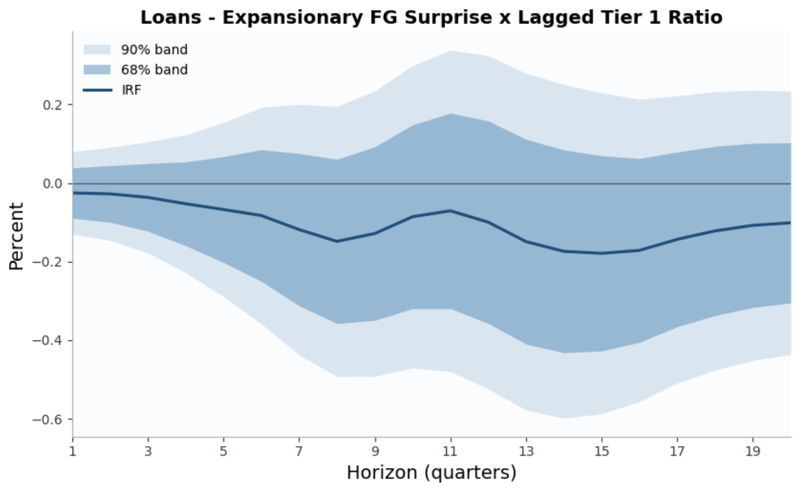


Figure 16: Average Interaction Effect: Expansionary Shock

Notes: The figure shows the impulse response of the interaction between expansionary forward-guidance shocks and banks' lagged Tier 1 capital ratios. For each horizon $h = 1, \ldots, 20$, I estimate the local projection

$$\begin{aligned}\Delta_h y_{i,t} = {} & \alpha_i^{(h)} + \lambda_t^{(h)} + \beta_{C,h}\left(FG_t^C \times Z_{i,t-1}\right) + \beta_{E,h}\left(FG_t^E \times Z_{i,t-1}\right) \\ & + \sum_{j=1}^{p} \Gamma'_{j,h} X_{i,t-j} \\ & + \sum_{j=1}^{p} \delta_{C,j,h}\left(FG_{t-j}^C \times Z_{i,t-j-1}\right) \\ & + \sum_{j=1}^{p} \delta_{E,j,h}\left(FG_{t-j}^E \times Z_{i,t-j-1}\right) + \varepsilon_{i,t+h}^{(h)}.\end{aligned}$$

The dependent variable is the cumulative loan response, $\Delta_\tau y_{i,t+\tau}$. The terms $\alpha_i^{(h)}$ and $\lambda_t^{(h)}$ denote bank and quarter fixed effects, respectively. The capitalization variable is the standardized lagged Tier 1 capital ratio,

$$Z_{i,t-1} = \frac{\text{Tier1Ratio}_{i,t-1} - \overline{\text{Tier1Ratio}_{t-1}}}{\text{sd}(\text{Tier1Ratio}_{t-1})}.$$

The vector $X_{i,t-j}$ contains lagged bank-level controls: log loans, deposits-to-assets, risk-weighted-assets-to-assets, accumulated other comprehensive income-to-equity, and the Tier 1 capital ratio. The lag length p is selected by AIC from candidate lag orders $p = 0, \ldots, 8$. Standard errors are clustered at the bank level. Figure 16 reports the differential response associated with expansionary forward-guidance shocks for banks with higher lagged Tier 1 exposure. The inner and outer bands are 68 and 90 percent confidence intervals, respectively.

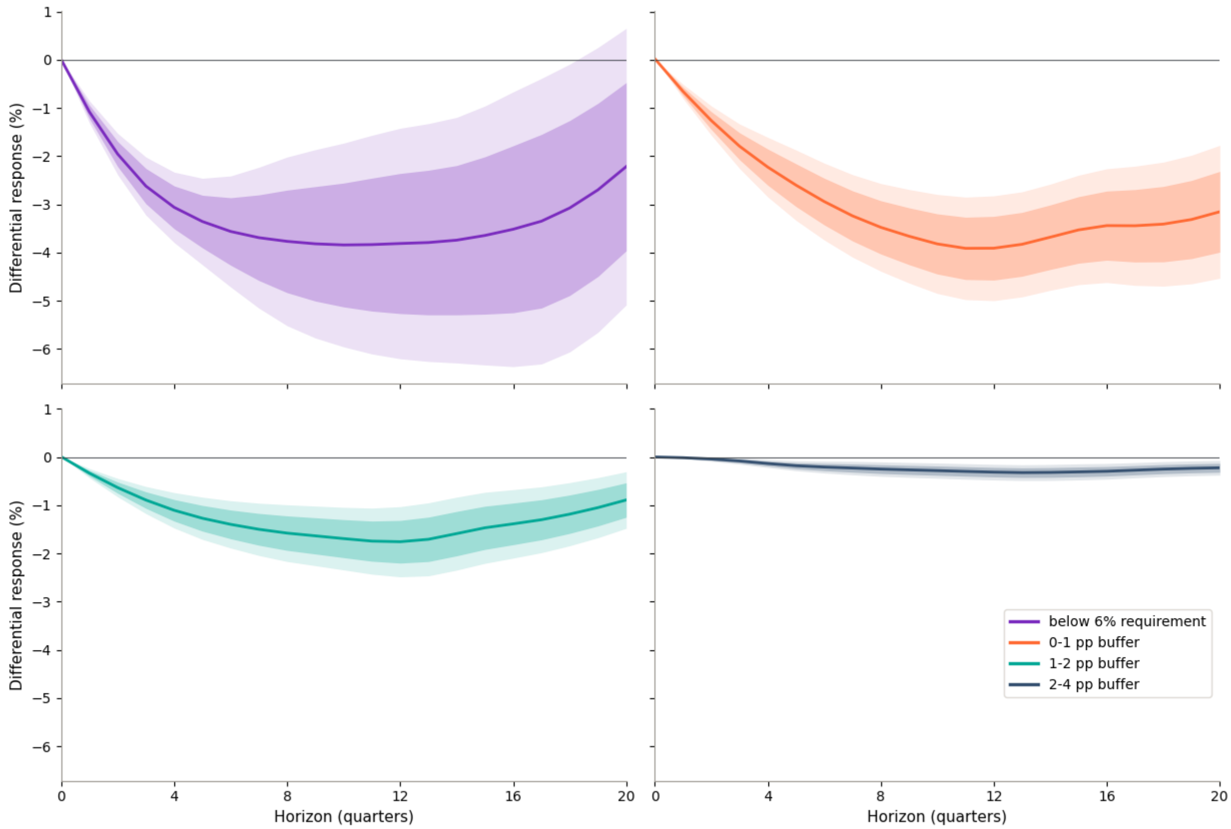


Figure 17: Lending Response to Contractionary FG by Bank Capitalization

The figure shows the interaction effect of a contractionary forward guidance shock with bank capitalization. The inner band corresponds to the 68% confidence band, while the outer band corresponds to the 90% confidence band. Each pane represents its counterpart in Figure 4. This figure provides the error bands promised in the main text.

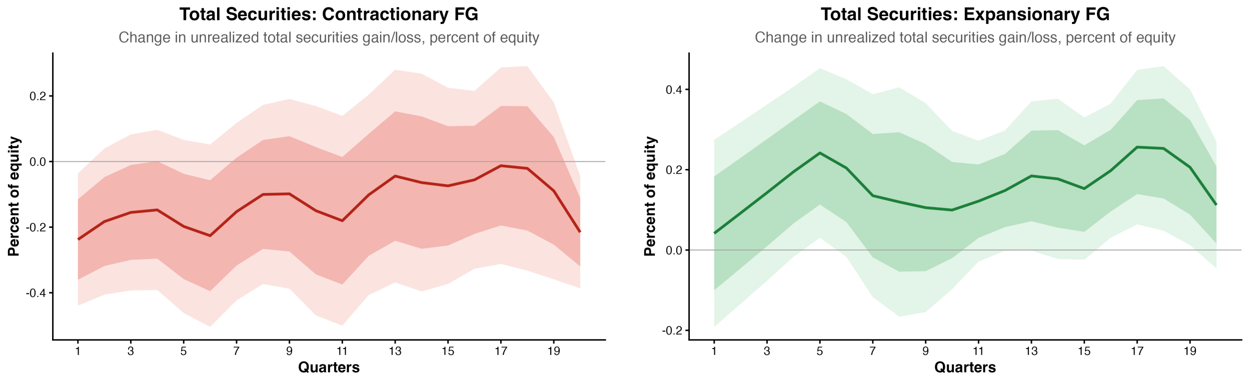


Figure 18: Bank Securities Responses to Forward Guidance Shocks

The figure shows the impulse response of total securities held by banks as a fraction of equity to a one-standard-deviation contractionary/expansionary forward-guidance shock. The inner dashed band corresponds to the 68% confidence band, while the outer solid band corresponds to the 90% confidence band.

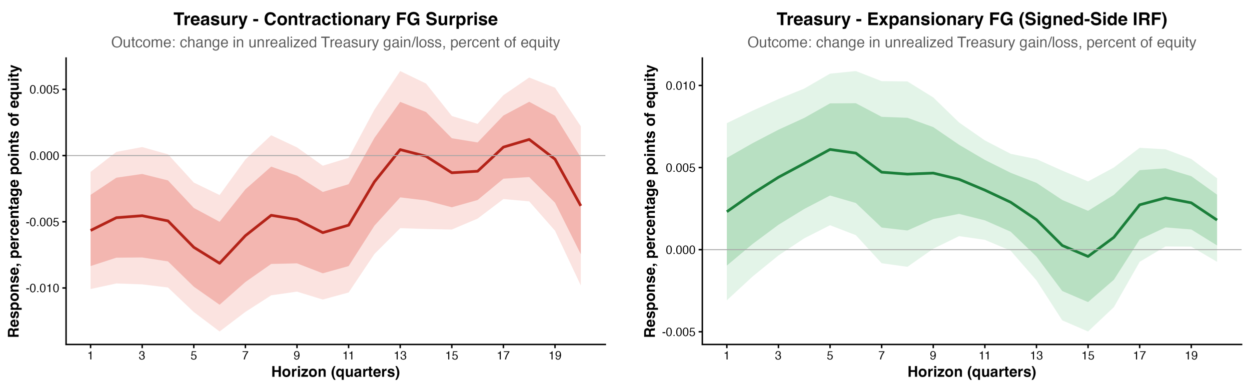


Figure 19: Bank Treasuries Responses to Forward Guidance Shocks

The figure shows the impulse response of total securities held by banks as a fraction of equity to a one-standard-deviation contractionary/expansionary forward-guidance shock. The inner dashed band corresponds to the 68% confidence band, while the outer solid band corresponds to the 90% confidence band.

Both Figure 19 and Figure 20 serve as attempts to decompose the evolution of total commercial bank securities as a fraction of equity after a contractionary forward guidance shock. Neither treasury holdings, nor residential MBS explain much of the movements in securities. I leave that exposition for another project.

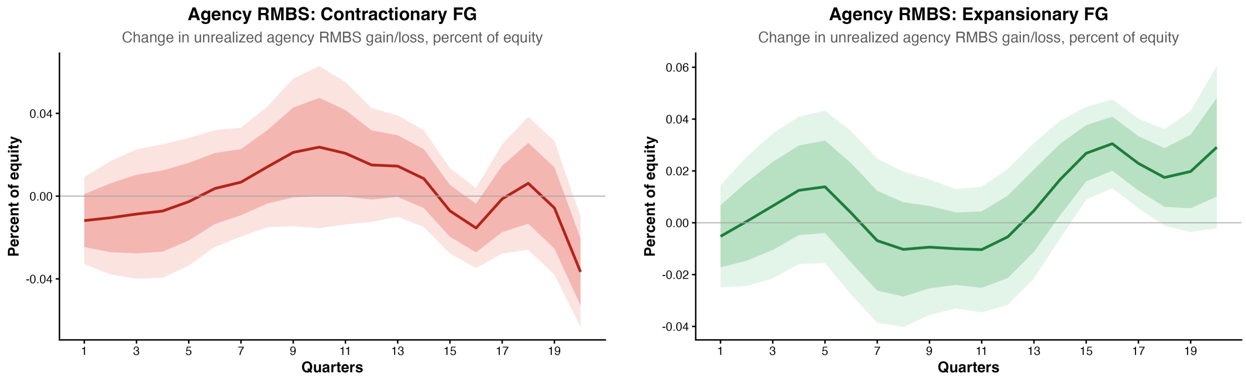


Figure 20: Commercial Banks' RMBS after FG Shock

Model

Household's Problem

The representative household maximizes lifetime utility:

$$\max_{\{C_t,N_t,B_{t+1},D_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{(C_t - \varphi H_t)^{1-\sigma}}{1-\sigma} - \theta \frac{N_t^{1+\eta}}{1+\eta} + \phi \ln\left(\frac{D_t + B_t^T}{P_t} \right) \right) \tag{A.1}$$

subject to the budget constraint:

$$P_t C_t + B_{t+1} + D_t + B_t^T \leq W_t N_t + \Psi_t + \Phi_t + (1+i_{t-1}^B) B_t + (1+i_{t-1}^D)(D_{t-1} + B_{t-1}^T) - P_t T_t \tag{A.2}$$

Lagrangian Formulation

Define the Lagrangian:

$$\begin{aligned} \mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Bigg[& \frac{(C_t - \varphi H_t)^{1-\sigma}}{1-\sigma} - \theta \frac{N_t^{1+\eta}}{1+\eta} + \phi \ln\left(\frac{D_t + B_t^T}{P_t} \right) \\ & + \lambda_t \left(W_t N_t + \Psi_t + \Phi_t + (1+i_{t-1}^B) B_t + (1+i_{t-1}^D)(D_{t-1} + B_{t-1}^T) - P_t T_t - P_t C_t - B_{t+1} - D_t - B_t^T \right) \Bigg] \end{aligned} \tag{A.3}$$

Final Goods Sector

The final goods sector combines a continuum of differentiated intermediate goods $Y_t(j)$ according to a Dixit-Stiglitz aggregator:

$$Y_t = \left(\int_0^1 Y_t(j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}}, \quad \epsilon > 1 \tag{A.4}$$

where ϵ represents the elasticity of substitution between intermediate goods.

Profit Maximization Problem

The final goods firm takes P_t as given and chooses $Y_t(j)$ to maximize profits:

$$\max_{\{Y_t(j)\}} P_t \left(\int_0^1 Y_t(j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}} - \int_0^1 P_t(j) Y_t(j) dj \tag{A.5}$$

Aggregate Price Level

The aggregate price level P_t is given by:

$$P_t = \left(\int_0^1 P_t(j)^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}} \tag{A.6}$$

Intermediate Goods Firm's Problem

Each intermediate goods producer $j \in [0,1]$ operates a constant returns to scale production function:

$$Y_t(j) = A_t N_t(j), \tag{A.7}$$

where A_t is the aggregate productivity shock.

The firm's profit function is given by:

$$\Psi_t(j) = P_t(j) Y_t(j) - (1 + i_t^L)(1 - \phi_{d,t+1}) W_t N_t(j). \tag{A.8}$$

Where firms finance their inputs via loans from the bank:

$$W_t N_t(j) = L_t(j) \tag{A.9}$$

So, we get:

$$\Psi_t(j) = P_t(j)Y_t(j) - (1 + i_t^L(j))(1 - \phi_{d,t+1}(j))L_t(j). \tag{A.10}$$

which allows us to endogenize default:

$$\phi_{d,t+1} = \max\left(1 - \frac{Y_t(j)P_t(j)}{(1 + i_t^L(j))L_t(j)}, 0\right) \tag{A.11}$$

which in aggregated terms is:

$$\phi_{d,t+1} = \max\left(1 - \frac{Y_t}{(1 + i_t^L)\ell_t}, 0\right) \tag{A.12}$$

Substituting for $N_t(j)$

Using the production function, we substitute $N_t(j) = \frac{Y_t(j)}{A_t}$ into the profit function:

$$\Psi_t(j) = P_t(j)Y_t(j) - (1 + i_t^L)(1 + \phi_{t+1})\frac{W_t}{A_t}Y_t(j). \tag{A.13}$$

Demand Constraint

The firm's demand function is derived from the final goods sector:

$$Y_t(j) = \left(\frac{P_t(j)}{P_t}\right)^{-\epsilon} Y_t. \tag{A.14}$$

Profit Function with Demand Substituted

Substituting $Y_t(j) = \left(\frac{P_t(j)}{P_t}\right)^{-\epsilon} Y_t$ into the profit function:

$$\Psi_t(j) = P_t(j)\left(\frac{P_t(j)}{P_t}\right)^{-\epsilon} Y_t - (1 + i_t^L)(1 + \phi_{t+1})\frac{W_t}{A_t}\left(\frac{P_t(j)}{P_t}\right)^{-\epsilon} Y_t. \tag{A.15}$$

Factor out $\left(\frac{P_t(j)}{P_t}\right)^{-\epsilon} Y_t$:

$$\Psi_t(j) = \left(\frac{P_t(j)}{P_t}\right)^{-\epsilon} Y_t \left[P_t(j) - (1 + i_t^L)(1 + \phi_{t+1})\frac{W_t}{A_t}\right]. \tag{A.16}$$

Profit Maximization Problem

Each intermediate goods firm chooses $P_t(j)$ to maximize:

$$\max_{P_t(j)} \quad \left(\frac{P_t(j)}{P_t}\right)^{-\epsilon} Y_t \left[P_t(j) - (1 + i_t^L)(1 + \phi_{t+1})\frac{W_t}{A_t}\right]. \tag{A.17}$$

Sticky Prices: Calvo Pricing

We assume price rigidity following Calvo (1983), where intermediate firms can reset prices with probability $1 - \xi$, and keep the previous price with probability ξ.

Firm's Pricing Problem

Each intermediate firm j maximizes expected discounted profits:

$$\max_{P_t(j)} X_t \sum_{s=0}^{\infty} \xi^s \Lambda_{t,t+s} \left[\frac{P_t(j)}{P_{t+s}} \left(\frac{P_t(j)}{P_{t+s}}\right)^{-\epsilon} Y_{t+s} - (1 + i_t^L)(1 + \phi_{t+1})\frac{w_{t+s}}{A_{t+s}} \left(\frac{P_t(j)}{P_{t+s}}\right)^{-\epsilon} Y_{t+s}\right]. \tag{A.18}$$

subject to the demand function:

$$Y_{t+s}(j) = \left(\frac{P_t(j)}{P_{t+s}}\right)^{-\epsilon} Y_{t+s}. \tag{A.19}$$

Reset Price under Calvo Pricing

Under the Calvo pricing assumption, a firm that updates its price at time t chooses $P_t(j)$ to maximize expected discounted profits. The optimal price-setting equation is:

$$P_t(j) = \frac{\epsilon}{\epsilon - 1} \frac{X_t \sum_{s=0}^{\infty} \xi^s \Lambda_{t,t+s} mc_{t+s} P_{t+s}^{\epsilon} Y_{t+s}}{X_t \sum_{s=0}^{\infty} \xi^s \Lambda_{t,t+s} P_{t+s}^{\epsilon-1} Y_{t+s}}. \tag{A.20}$$

Since nothing on the right-hand side depends on j, all updating firms choose the same price. We define this common reset price as $P_t^{\#}$, leading to:

$$P_t^{\#} = \frac{\epsilon}{\epsilon - 1} \frac{X_{1,t}}{X_{2,t}}, \tag{A.21}$$

where the two auxiliary variables $X_{1,t}$ and $X_{2,t}$ evolve recursively as:

$$X_{1,t} = mc_t P_t^{\epsilon} Y_t + \xi X_t \Lambda_{t,t+1} X_{1,t+1}, \tag{A.22}$$

$$X_{2,t} = P_t^{\epsilon-1} Y_t + \xi X_t \Lambda_{t,t+1} X_{2,t+1}. \tag{A.23}$$

Key Properties

- Since the right-hand side does not depend on j, all updating firms set the same **reset price** $P_t^{\#}$.
- $X_{1,t}$ represents the expected future discounted marginal cost-weighted demand.
- $X_{2,t}$ represents the expected future discounted nominal demand.

Real Marginal Cost

The firm's real marginal cost is given by:

$$mc_t = (1 + i_t^L)(1 - \phi_{t+1}) \frac{w_t}{A_t}. \tag{A.24}$$

Sticky Wages

We have a labor unions of unit mass, indexed by $i \in [0, 1]$. The unions employ household labor and remunerate them at a rate w_t. The unions then turnaround and sell this labor to a "labor packer" at a price of $U_t(h)$. This labor is then transformed by the packer into a a final labor item that firms can employ in the form of a CES technology:

$$N_{d,t} = \left[\int_0^1 N_t(i)^{\frac{\epsilon_w - 1}{\epsilon_w}} di \right]^{\frac{\epsilon_w}{\epsilon_w - 1}} \tag{A.25}$$

We end up with the following wage index and individual union labor demand after profit maximizaiton:

$$U_t^{1-\epsilon_w} = \int_0^1 U_t(i)^{1-\epsilon_w} di$$
$$N_t(i) = \left(\frac{U_t(i)}{U_t} \right)^{-\epsilon_w} N_{d,t}$$

With probability $1 - \phi_w$, the unions can adjust the wage. They face the following problem:

$$\max_{N_t(i)} E_t \sum_{j=0}^{\infty} \phi_w^j \Lambda_{t,t+j} \left\{ U_t(i)^{1-\epsilon_w} U_{t+j}^{\epsilon_w} P_{t+j}^{-1} N_{d,t+j} - w_{t+j} U_t(i)^{-\epsilon_w} U_{t+j}^{\epsilon_w} N_{d,t+j} \right\}$$

After taking the FOC, we can rearrange to get the optimal reset wage $U_t^{\#}$:

$$U_t^{\#} = \frac{\epsilon_w}{\epsilon_w - 1} \frac{E_t \sum_{j=0}^{\infty} \phi_w^j \Lambda_{t,t+j} w_{t+j} U_{t+j}^{\epsilon_w} N_{d,t+j}}{E_t \sum_{j=0}^{\infty} \phi_w^j \Lambda_{t,t+j} P_{t+j}^{-1} U_{t+j}^{\epsilon_w} N_{d,t+j}}$$

Defining $f_{1,t} = \frac{F_{1,t}}{P_t^{\epsilon_w}}$ and $f_{2,t} = \frac{F_{2,t}}{P_t^{\epsilon_w - 1}}$, we derive:

$$u_t^{\#} = \frac{\epsilon_w}{\epsilon_w - 1} \frac{f_{1,t}}{f_{2,t}} \tag{A.26}$$

Where we define $f_{1,t}$ and $f_{2,t}$ recursively:

$$f_{1,t} = w_t u_t^{\epsilon_w} N_{d,t} + \phi_w E_t \Lambda_{t,t+1} \Pi_{t+1}^{\epsilon_w} f_{1,t+1} \tag{A.27}$$

$$f_{2,t} = u_t^{\epsilon_w} N_{d,t} + \phi_w E_t \Lambda_{t,t+1} \Pi_{t+1}^{\epsilon_w - 1} f_{2,t+1} \tag{A.28}$$

Aggregate real wage evolution is expressed as follows (which follows a similar derivation as the inflation rate evolution):

$$u_t^{1-\epsilon_w} = (1-\phi_w)(u_t^{\#})^{1-\epsilon_w} + \phi_w \Pi_t^{\epsilon_w - 1} u_{t-1}^{1-\epsilon_w} \tag{A.29}$$

Since labor supply must equal that demanded by the unions:

$$N_t = \int_0^1 N_{d,t}(i)di$$

We end up with:

$$N_t = N_{d,t} v_t^w$$

With v_t^w representing wage dispersion:

$$v_t^w = (1-\phi_w)\left(\frac{u_t^{\#}}{u_t}\right)^{-\epsilon_w} + \phi_w \Pi_t^{\epsilon_w}\left(\frac{u_t}{u_{t-1}}\right)^{\epsilon_w} v_{t-1}^w$$

B Bank's Problem

The bank chooses $\{L_t, B_t, R_t, D_t, X_t\}$ to maximize:

$$\Phi_t = E_t\Big\{\Lambda_{t+1}\Big[(1+i_t^L)(1-\phi_d)L_t + (1+i_t^B)B_t + (1+i_t^R)R_t - (1+i_t^D)D_t\Big]\Big\} - X_t - \frac{\phi_x}{2}(L_t - L_{t-1})^2 \tag{B.1}$$

subject to:

$$L_t + B_t + R_t = D_t + (1 - f(\delta_t))X_t, \quad \text{where} \quad \delta_t = \frac{L_t}{X_t}, \tag{B.2}$$

$$f(\delta_t) = \frac{\alpha}{2}\delta_t^2, \tag{B.3}$$

$$R_t \geq \rho D_t, \quad 0 \leq \rho < 1. \tag{B.4}$$

C Lagrangian Formulation

Defining multipliers:

- μ_t for the balance-sheet constraint,
- $\alpha_t \geq 0$ for the reserve requirement.

The Lagrangian is:

$$\mathcal{L} = E_t\Big[\Lambda_{t+1}\big((1+i_t^L)(1-\phi_d)L_t + (1+i_t^B)B_t + (1+i_t^R)R_t - (1+i_t^D)D_t\big)\Big]$$
$$-X_t - \frac{\phi_x}{2}(L_t - L_{t-1})^2 - \mu_t\Big(L_t + B_t + R_t - D_t - \Big(1 - \frac{\alpha}{2}\delta_t^2\Big)X_t\Big) - \alpha_t(R_t - \rho D_t).$$

D Loan - Bond Spread Condition

Given our FOC conditions, we can derive the interest rate spread between loans and bonds. Substitute the value of μ_t from (44) into the FOC for L:

$$\Lambda_{t+1}(1+i_t^L)(1-\phi_d) - \phi_x\big(L_t - L_{t-1}\big) - \Lambda_{t+1}(1+i_t^B)\Big[1 + \alpha\,\delta_t\Big] = 0. \tag{D.1}$$

Dividing the entire equation by Λ_{t+1} (which is positive) yields

$$(1+i_t^L)(1-\phi_d) = (1+i_t^B)(1+\alpha\,\delta_t) + \frac{\phi_x}{\Lambda_{t+1}}\big(L_t - L_{t-1}\big). \tag{D.2}$$

I wish to express the gross spread $\frac{1+i_t^L}{1+i_t^B}$. Divide both sides of (2) by $1+i_t^B$:

$$\frac{1+i_t^L}{1+i_t^B}(1-\phi_d) = 1 + \alpha\,\delta_t + \frac{\phi_x}{\Lambda_{t+1}(1+i_t^B)}\big(L_t - L_{t-1}\big). \tag{D.3}$$

Finally, divide both sides by $1-\phi_d$ to obtain:

$$\frac{1+i_t^L}{1+i_t^B} = \frac{1+\alpha\,\delta_t}{1-\phi_d} + \frac{\phi_x}{\Lambda_{t+1}(1+i_t^B)(1-\phi_d)}\big(L_t - L_{t-1}\big). \tag{D.4}$$

Final Result

Equation (53) is the stand-alone interest rate spread condition that results from the bank's optimization problem with adjustment costs:

$$\frac{1+i_t^L}{1+i_t^B} = \frac{1+\alpha\,\delta_t}{1-\phi_d} + \frac{\phi_x}{\Lambda_{t+1}(1+i_t^B)(1-\phi_d)}(L_t - L_{t-1})$$

This equation shows that, in addition to the baseline spread $\frac{1+\alpha\,\delta_t}{1-\phi_d}$ (which would arise in the absence of adjustment costs), there is an extra term that accounts for the cost of adjusting the loan portfolio.

E Equity

My specification for the evolution of bank equity is based on that employed by Benigno and Benigno (2021). Equity has the following specification:

$$x_t = \left[1 - \frac{(1+i_t^L)}{(1+i_t^B)}(1-\phi_{d,t+1})\right]\ell_t + \frac{i_t^D - i_t^B}{1+i_t^B}d_t + \frac{i_t^B - i_t^R}{1+i_t^B}r_t \tag{E.1}$$

F Treasury Bond Issuance and Tax Rule

Let there be a central bank with the following budget constraint:

$$B_t^T = (1+i_t^D)B_{t-1}^T - T_t \tag{F.1}$$

Further, assume that the government sets the tax policy in the following manner:

$$T_t = \tau Y_t, \qquad 0 < \tau < 1 \tag{F.2}$$

G Taylor Rule (Monetary Policy)

The central bank follows a Taylor Rule, adjusting the nominal interest rate in response to inflation and output deviations:

$$1+i_t^R = (1+i^R)^\rho \left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi} \left(\frac{Y_t}{Y^*}\right)^{\phi_Y} e^{\phi_t}. \tag{G.1}$$

Where i^R, Π^*, and Y^* are the steady state values of the interest rate on reserve balances, inflation, and output, respectively. Shocks to the interest rate on reserves enter in through ϕ_t.

The shock to the policy rate is specified as:

$$\phi_t = \sum_{i=0}^{20} \psi^i_{r,t-i} \tag{G.2}$$

w_t is a weight that decays geometrically over time.

Let $\hat{\psi}_{r,t}$ denote one's belief (or prior) about the true value of $\psi_{r,t}$, which represents the true policy innovation at time t. $\hat{\psi}_{r,t}$ is defined as follows:

$$\hat{\psi}^j_{r,t} = \hat{\psi}^j_{r,t-1} + \kappa_j(s^j_t - \hat{\psi}^j_{r,t-1}), \quad j = 1, ..., 20. \tag{G.3}$$

Here, κ_j represents the Kalman gain with respect to horizon j, and $s^j_t - \hat{\psi}^j_{r,t-1}$ is the forecast error.

The forward guidance shock ψ^{FG}_t is defined as a set of signals:

$$\psi^{FG}_t = [s^0_t \;\, s^1_t \dots s^{20}_t]$$

The current policy rate is perfectly observed:

$$s^0_t = \psi^0_{r,t}$$

While future deviations are unobserved, i.e. for $j \geq 1$

$$s^j_t = \psi^j_{r,t} + \nu_t, \qquad \nu_t \sim N(0, \sigma^2_\nu) \tag{G.4}$$

Hence, the Kalman gain $\kappa = \frac{\sigma^2_\psi}{\sigma^2_\psi + \sigma^2_\nu}$, where σ^2_ψ is the variance of the agent's prior.

H Equilibrium Conditions

$$E_t\{\Lambda_{t+1}(1+i_t^B)\} = 1 \tag{H.1}$$

$$\theta N_t^{\eta} = (C_t - \varphi H_t)^{-\sigma} w_t \tag{H.2}$$

$$d_t = \phi (C_t - \varphi H_t)^{\sigma} \frac{(1+i_t^B)}{(i_t^B - i_t^D)} - b_t^T \tag{H.3}$$

$$\Lambda_t = \beta \left(\frac{C_t - \varphi H_t}{C_{t-1} - \varphi H_{t-1}} \right)^{-\sigma} \Pi_t^{-1} \tag{H.4}$$

$$1 = (1-\xi)(p_t^{\#})^{1-\epsilon} + \xi \Pi_t^{\epsilon-1} \tag{H.5}$$

$$u_t^{1-\epsilon_w} = (1-\phi_w)(u_t^{\#})^{1-\epsilon_w} + \phi_w \Pi_t^{\epsilon_w - 1} u_{t-1}^{1-\epsilon_w} \tag{H.6}$$

$$p_t^{\#} = \frac{\epsilon}{\epsilon - 1} \frac{\hat{X}_{1,t}}{\hat{X}_{2,t}} \tag{H.7}$$

$$u_t^{\#} = \frac{\epsilon_w}{\epsilon_w - 1} \frac{f_{1,t}}{f_{2,t}} \tag{H.8}$$

$$\hat{X}_{1,t} = mc_t Y_t + \xi E_t \Lambda_{t+1} \Pi_{t+1}^{\epsilon} \hat{X}_{1,t+1} \tag{H.9}$$

$$\hat{X}_{2,t} = Y_t + \xi E_t \Lambda_{t+1} \Pi_{t+1}^{\epsilon-1} \hat{X}_{2,t+1} \tag{H.10}$$

$$f_{1,t} = w_t u_t^{\epsilon_w} N_{d,t} + \phi_w E_t \Lambda_{t+1} \Pi_{t+1}^{\epsilon_w} f_{1,t+1} \tag{H.11}$$

$$f_{2,t} = u_t^{\epsilon_w} N_{d,t} + \phi_w E_t \Lambda_{t+1} \Pi_{t+1}^{\epsilon_w - 1} f_{2,t+1} \tag{H.12}$$

$$u_t = mc_t A_t \tag{H.13}$$

$$mc_t = (1+i_t^L)(1-\phi_{d,t+1}) \frac{w_t}{A_t} \tag{H.14}$$

$$Y_t = C_t + \frac{\alpha}{2} \delta_t^2 X_t + \frac{\varphi_x}{2}(L_t - L_{t-1})^2 \tag{H.15}$$

$$b_t^T = (1+i_t^D) b_{t-1}^T - T_t \tag{H.16}$$

$$T_t = \tau Y_t \tag{H.17}$$

$$A_t N_t = Y_t \nu_t^P \tag{H.18}$$

$$N_t = N_{d,t} \nu_t^w \tag{H.19}$$

$$\nu_t^P = (1-\xi)(p_t^{\#})^{-\epsilon} + \xi \Pi_t^{\epsilon} \nu_{t-1}^P \tag{H.20}$$

$$\nu_t^w = (1-\phi_w) \left(\frac{u_t^{\#}}{u_t} \right)^{-\epsilon_w} + \phi_w \Pi_t^{\epsilon_w} \left(\frac{u_t}{u_{t-1}} \right)^{\epsilon_w} \nu_{t-1}^w \tag{H.21}$$

$$\ln A_t = \rho_A \ln A_{t-1} + s_A \epsilon_{A,t} \tag{H.22}$$

$$1 + i_t^R = (1 + i^R)^{\rho_r} \left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi} \left(\frac{Y_t}{Y^*}\right)^{\phi_Y} e^{\epsilon_t^{FG}} \tag{H.23}$$

$$\phi_{d,t+1} = 1 - \frac{Y_t}{(1 + i_t^L)\ell_t} \tag{H.24}$$

$$\frac{(1 + i_t^L)}{(1 + i_t^B)} = \frac{1 + \alpha\delta_t}{1 - \phi_{d,t+1}} + \frac{\phi_x}{\Lambda_{t+1}(1 + i_t^B)(1 - \phi_{d,t+1})}(\ell_t - \ell_{t-1}) \tag{H.25}$$

$$\ell_t = \rho_\ell \ell_{t-1} + (1 - \rho_\ell)\left(\left(\frac{1}{\rho} - 1\right) r_t + \left(1 - \frac{\alpha}{2}\delta_t^2\right) x_t - b_t\right) \tag{H.26}$$

$$r_t = \frac{1}{1 - \rho}\left[\left(1 - \frac{\alpha}{2}\delta_t^2\right) x_t - \left(\frac{(1 + i_t^L) + (1 + i_t^B)}{(1 + i_t^L)}\right) w_t N_t\right] \tag{H.27}$$

$$\ell_t + b_t + r_t = d_t + \left(1 - \frac{\alpha}{2}\delta_t^2\right) x_t \tag{H.28}$$

$$x_t = \left[1 - \frac{(1 + i_t^L)}{(1 + i_t^B)}(1 - \phi_{d,t+1})\right] \ell_t + \frac{i_t^D - i_t^B}{1 + i_t^B} d_t + \frac{i_t^B - i_t^R}{1 + i_t^B} r_t \tag{H.29}$$

$$\delta_t = \frac{\ell_t}{x_t} \tag{H.30}$$

Steady State

In deriving the Steady State, I normalize the following set of variables equal to 1:

$$Y = C = \Pi = 1$$

The steady state stochastic discount factor becomes:

$$\Lambda = \beta$$

From which it follows that our interest rate on private debt is:

$$i_B = \frac{1}{\beta} - 1$$

Since any variable $x_t = x_{t-1} = x$ in steady state, we have the following:

$$\ell = \left(\frac{1}{\rho} - 1\right) r + \left(1 - \frac{\alpha}{2}\delta^2\right) x - b$$

$$1 + i_L = \frac{1 + i_B}{1 - \phi_d}$$
$$\ell = \frac{1}{1 + i_B}$$

Adding an additional normalization in the form of $p^{\#} = 1$, we get:

$$mc = \frac{\epsilon - 1}{\epsilon}$$

Which follows from the fact that:

$$X1 = \frac{mc}{1 - \xi\beta}$$
$$X2 = \frac{1}{1 - \xi\beta}$$

Which comes fromt the fact that $\Lambda = \beta$, $\Pi = 1$, and $X_{1,t} = X_{1,t+1} = X_1$ in steady state. The remaining conditions are as follows:

$$\theta = w \tag{H.31}$$
$$\nu^P = 1 \tag{H.32}$$
$$mc = (1 + i_L)(1 - \phi_d)\frac{w}{A} \tag{H.33}$$
$$\phi_d = 1 - \frac{Y}{(1 + i_L)\ell} \tag{H.34}$$
$$i_D = \frac{\tau}{b_T} \tag{H.35}$$
$$d = \varphi\frac{1 + i_B}{i_B - i_D} - b_T \tag{H.36}$$
$$\ell + b + r = d + \left(1 - \frac{\alpha}{2}\delta^2\right)x \tag{H.37}$$
$$x = \left[1 - \frac{(1 + i_L)}{(1 + i_B)}(1 - \phi_d)\right]\ell + \frac{i_D - i_B}{1 + i_B}d + \frac{i_B - i_R}{1 + i_B}r \tag{H.38}$$
$$A = 1 \tag{H.39}$$
$$f1 = \frac{w}{1 - \phi_w\beta} \tag{H.40}$$

$$f2 = \frac{1}{1-\phi_w \beta} \tag{H.41}$$

$$u^{\#} = 1 \tag{H.42}$$

$$\delta = \frac{l}{x} \tag{H.43}$$

Log Linearized Conditions

$$E_t \tilde{\Lambda}_{t+1} + \tilde{i}_{B,t} = 0 \tag{H.44}$$

$$\eta \tilde{N}_t - \tilde{W}_t + \sigma \frac{C^*}{C^* - \varphi H^*} \tilde{C}_t - \sigma \frac{\varphi H^*}{C^* - \varphi H^*} \tilde{H}_t = 0 \tag{H.45}$$

$$\begin{aligned} \tilde{d}_t = & \frac{\phi (C^* - \varphi H^*)^{\sigma-1} \sigma}{d^*} \frac{1 + i^{B*}}{i^{B*} - i^{D*}} (\tilde{C}_t - \tilde{H}_t) \\ & - \frac{\phi (C^* - \varphi H^*)^{\sigma}}{d^*} \frac{1 + i^{D*}}{(i^{B*} - i^{D*})^2} \tilde{i}_t^B + \frac{\phi (C^* - \varphi H^*)^{\sigma}}{d^*} \frac{1 + i^{B*}}{(i^{B*} - i^{D*})^2} \tilde{i}_t^D - \frac{b^{T*}}{d^*} \tilde{b}_t^T \end{aligned} \tag{H.46}$$

$$\tilde{\Lambda}_t = -\sigma \frac{C^*}{C^* - \varphi H^*} (\tilde{C}_t - \tilde{C}_{t-1}) - \sigma \frac{\varphi H^*}{C^* - \varphi H^*} (\tilde{H}_t - \tilde{H}_{t-1}) - \tilde{\Pi}_t \tag{H.47}$$

$$0 = (1-\xi)(1-\epsilon)(p^{\#*})^{1-\epsilon} \tilde{p}_t^{\#} + \xi(\epsilon - 1)(\Pi^*)^{\epsilon-1} \tilde{\Pi}_t \tag{H.48}$$

$$\tilde{u}_t = \frac{(1-\phi_w)) u^{\#1-\epsilon_w *}}{u^{*1-\epsilon_w}} \tilde{u}_t^{\#} + \phi_w (\epsilon_w - 1) \Pi^{*\epsilon_w - 1} (\tilde{\Pi}_t - \tilde{u}_{t-1}) \tag{H.49}$$

$$\tilde{p}_t^{\#} = \tilde{X}_{1,t} - \tilde{X}_{2,t} \tag{H.50}$$

$$\tilde{u}_t^{\#} = \tilde{f}_{1,t} - \tilde{f}_{2,t} \tag{H.51}$$

$$\tilde{X}_{1,t} = \frac{mc^* Y^*}{X_1^*} (\tilde{mc}_t + \tilde{Y}_t) + \xi \Lambda^* (\Pi^*)^{\epsilon} E_t (\tilde{\Lambda}_{t+1} + \epsilon \tilde{\Pi}_{t+1} + \tilde{X}_{1,t+1}) \tag{H.52}$$

$$\tilde{X}_{2,t} = \frac{Y^*}{X_2^*} \tilde{Y}_t + \xi \Lambda^* (\Pi^*)^{-1} E_t (\tilde{\Lambda}_{t+1} - \tilde{\Pi}_{t+1} + \tilde{X}_{2,t+1}) \tag{H.53}$$

$$\tilde{f}_{1,t} = \frac{w^* u^{\epsilon_w *} N_d^*}{f_1^*} (\tilde{w}_t + \epsilon_w \tilde{u}_t + \tilde{N}_{d,t}) + \phi_w \Lambda^* \Pi^{\epsilon_w *} (\tilde{\Lambda}_{t+1} + \epsilon_w \tilde{\Pi}_{t+1} + \tilde{f}_{1,t+1}) \tag{H.54}$$

$$\tilde{f}_{2,t} = \frac{u^{\epsilon_w *} N_d^*}{f_2^*} (\tilde{u}_t + \tilde{N}_{d,t}) + \phi_w \Lambda^* \Pi^{\epsilon_w - 1 *} (\tilde{\Lambda}_{t+1} + (\epsilon_w - 1) \tilde{\Pi}_{t+1} + \tilde{f}_{2,t+1}) \tag{H.55}$$

$$\tilde{u}_t = \tilde{mc}_t + \tilde{A}_t \tag{H.56}$$

$$\tilde{mc}_t = \tilde{i}_t^L - \frac{1}{1-\phi_d^*} \tilde{\phi}_{d,t+1} + \tilde{w}_t - \tilde{A}_t \tag{H.57}$$

$$\tilde{Y}_t = \frac{C^*}{Y^*} \tilde{C}_t + \frac{\alpha (\delta^*)^2 X^*}{Y^*} (\tilde{X}_t + \tilde{\delta}_t) \tag{H.58}$$

$$\tilde{b}^T_t = (1 + i^{D*})\tilde{b}^T_{t-1} + \tilde{i}^D_t - \frac{T^*}{b^{T*}}\tilde{T}_t \tag{H.59}$$

$$\tilde{T}_t = \tau\tilde{Y}_t \tag{H.60}$$

$$\tilde{A}_t + \tilde{N}_t = \tilde{Y}_t + \tilde{\nu}^P_t \tag{H.61}$$

$$\tilde{N}_t = \tilde{N}_{t,t} + \tilde{\nu}^w_t \tag{H.62}$$

$$\tilde{\nu}^P_t = \xi(\Pi^*)^\epsilon(\epsilon\tilde{\Pi}_t + \tilde{\nu}^P_{t-1}) - \frac{\epsilon(1-\xi)(p^{\#*})^{-\epsilon}}{\nu^{P*}}\tilde{p}^\#_t \tag{H.63}$$

$$\tilde{\nu}^w_t = \frac{(1-\phi_w)\epsilon_w\left(\frac{u^{\#*}}{u^*}\right)^{-\epsilon_w}}{\nu^{w*}}(\tilde{u} - \tilde{u}^\#_t) + \phi_w\Pi^{\epsilon_w}(\epsilon_w\tilde{\Pi}_t + \epsilon_w(\tilde{u}_t - \tilde{u}_{t-1}) + \tilde{\nu}^w_{t-1}) \tag{H.64}$$

$$\tilde{A}_t = \rho_A\tilde{A}_{t-1} + \gamma_A\varepsilon_{A,t} \tag{H.65}$$

$$\tilde{i}^R_t = (1 + i^{R*})^{\rho_r}(\phi_\pi\tilde{\Pi}_t + \phi_Y\tilde{Y}_t + \tilde{\varepsilon}^{FG}_t) \tag{H.66}$$

$$\tilde{\phi}_{d,t+1} = \frac{1-\phi^*_d}{\phi^*_d}(\tilde{i}^L_t + \tilde{\ell}_t - \tilde{Y}_t) \tag{H.67}$$

$$\frac{1}{1+i^{B*}}\tilde{i}^L_t - \frac{1+i^{L*}}{(1+i^{B*})^2}\tilde{i}^B_t = \frac{\alpha\delta^*}{1-\phi^*_d}\tilde{\delta}_t + \frac{(1+\alpha\delta^*)\phi^*_d}{(1-\phi^*_d)^2}\tilde{\phi}_{d,t+1} + \frac{\phi_x l^*}{\Lambda^*(1+i^{B*})(1-\phi^*_d)}(\tilde{l}_t - \tilde{l}_{t-1}) \tag{H.68}$$

$$\tilde{\ell}_t = \rho_\ell\tilde{\ell}_{t-1} + \frac{(1-\rho_\ell)}{\ell^*}\left[\left(\frac{1}{\rho} - 1\right)r^*\tilde{r}_t + \left(1 - \frac{\alpha}{2}(\delta^*)^2\right)x^*\tilde{x}_t - \alpha\delta^*(x^*)^2\tilde{\delta}_t - b^*\tilde{b}_t\right] \tag{H.69}$$

$$\ell^*\tilde{\ell}_t + b^*\tilde{b}_t + r^*\tilde{r}_t = d^*\tilde{d}_t + \left(1 - \frac{\alpha}{2}(\delta^*)^2\right)x^*\tilde{x}_t - \alpha\delta^*x^*\tilde{\delta}_t \tag{H.70}$$

$$\tilde{r}_t = \frac{1}{r^*}\left[\left(1 - \frac{\alpha}{2}\delta^{*2}\right)x^*\tilde{x}_t - \alpha\delta^{*2}x^*\tilde{\delta}_t - \frac{w^*N^*}{1+i^*_L}\tilde{i}^B_t + \frac{1+i^*_B}{(1+i^*_L)^2}w^*N^*\tilde{i}^L_t - \left(1 + \frac{1+i^*_B}{1+i^*_L}\right)w^*N^*(\tilde{w}_t + \tilde{N}_t)\right] \tag{H.71}$$

$$\begin{aligned}\tilde{x}_t = &\left[1 - \frac{1+i^*_L}{1+i^*_B}(1-\phi^*_d)\right]\frac{l^*}{x^*}\tilde{l}_t + \frac{i^*_D - i^*_B}{1+i^*_B}\frac{d^*}{x^*}\tilde{d}_t + \frac{i^*_B - i^*_R}{1+i^*_B}\frac{r^*}{x^*}\tilde{r}_t + \\ &\frac{l^*}{x^*}\left[-\frac{(1-\phi^*_d)}{(1+i^*_B)}\tilde{i}^L_t + \frac{(1+i^*_L)(1-\phi^*_d)}{(1+i^*_B)^2}\tilde{i}^B_t + \frac{(1+i^*_L)}{(1+i^*_B)}\tilde{\phi}_{d,t+1}\right] + \frac{d^*}{x^*}\left[\frac{1}{1+i^*_B}\tilde{i}^D_t - \frac{1+i^*_D - i^*_B}{(1+i^*_B)^2}\tilde{i}^B_t\right] + \\ &\frac{r^*}{x^*}\left[\frac{1+i^*_R - i^*_B}{(1+i^*_B)^2}\tilde{i}^B_t - \frac{1}{1+i^*_B}\tilde{i}^R_t\right]\end{aligned} \tag{H.72}$$

$$\tilde{\delta}_t = \tilde{\ell}_t - \tilde{x}_t \tag{H.73}$$

Table 6: Prior and Posterior Distribution of the Structural Parameters

Parameter	**Prior distribution**		**Posterior distribution**		
	Dist.	Mean / Std. dev.	Mean	5%	95%
Model Parameters					
ρ_A	Beta	0.800 / 0.150	0.8590	0.8022	0.9201
ϕ_π	Normal	1.500 / 1.000	1.3405	0.3804	2.3038
ϕ_y	Normal	0.500 / 0.600	0.3155	−1.3936	2.0172
ξ	Beta	0.550 / 0.250	0.7824	0.7189	0.8538
ϕ_w	Beta	0.750 / 0.150	0.4930	0.4535	0.5353
ρ_b	Beta	0.850 / 0.080	0.9426	0.9036	0.9838
ϕ_x	Gamma	1.200 / 0.600	0.3670	0.0939	0.6319
γ_b	Gamma	0.100 / 0.050	0.1372	0.0718	0.2011
λ_h	Beta	0.900 / 0.050	0.9554	0.9274	0.9848
ε	Normal	6.000 / 2.000	6.0186	2.7377	9.2301
ε_w	Normal	4.500 / 1.000	3.1864	2.3972	4.1106
Standard deviations of shocks					
σ_l	Gamma	0.150 / 0.060	0.1606	0.1276	0.1941
σ_d	Gamma	0.150 / 0.060	0.3878	0.3392	0.4366
σ_ϕ	Gamma	0.120 / 0.050	0.0350	0.0178	0.0517
σ_A	Inv. Gamma	0.300 / 0.050	0.2151	0.1903	0.2388
σ_{FG}	Inv. Gamma	0.100 / 0.010	0.1315	0.1177	0.1453
σ_y	Inv. Gamma	0.200 / 0.080	0.2274	0.1851	0.2689
σ_c	Inv. Gamma	0.200 / 0.080	0.1822	0.1632	0.2009
σ_π	Inv. Gamma	2.000 / 0.250	2.2134	2.0055	2.4167
σ_{iL}	Inv. Gamma	2.500 / 0.500	2.3492	2.1166	2.5724
σ_{iD}	Inv. Gamma	3.000 / 0.500	2.5530	2.3001	2.8017
σ_b	Inv. Gamma	0.004 / 0.002	0.0047	0.0025	0.0070
σ_{π_w}	Inv. Gamma	0.180 / 0.070	0.1337	0.1173	0.1498
σ_{iR}	Inv. Gamma	0.180 / 0.080	0.1042	0.0740	0.1335

Estimated Parameters

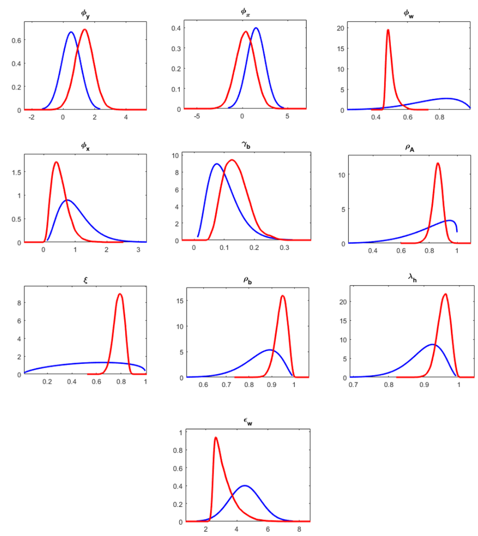


Figure 21: Prior and Posterior Marginal Distributions of Parameters

The posterior densities were generates from four chains with 250,000 draws each using the Metropolis algorithm. Red lines denote posterior distributions. Blue lines denote prior distributions

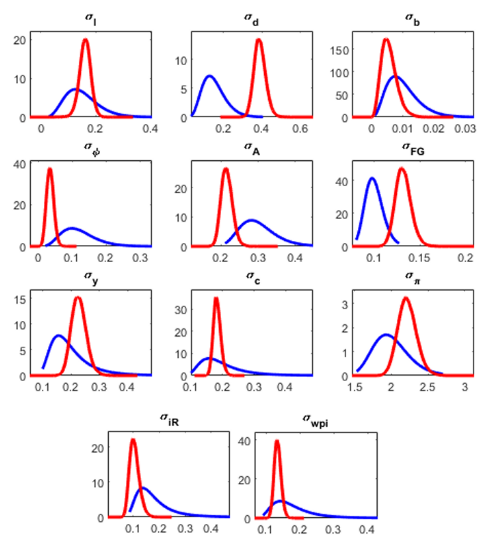
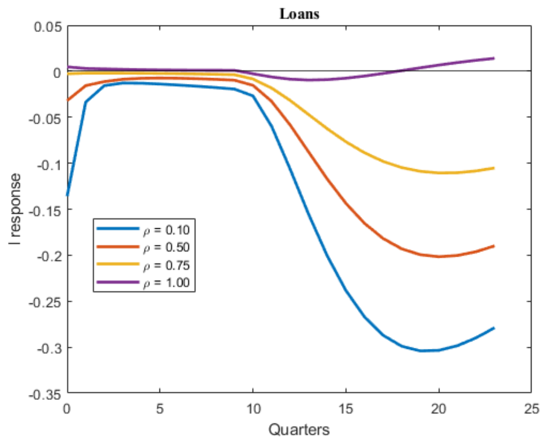


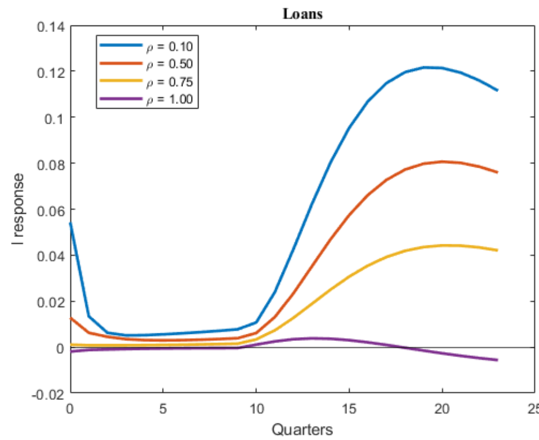
Figure 22: Prior and Posterior Marginal Distributions of Shocks

The posterior densities were generates from four chains with 250,000 draws each using the Metropolis algorithm. Red lines denote posterior distributions. Blue lines denote prior distributions

Additional Model Plots



(a) Contractionary FG Surprise



(b) Expansionary FG Surprise

Figure 23: Responses of loans under different reserve ratios

ρ denotes the reserve requirement. When $\rho = 0.10$, banks must hold 10% of deposits in reserves; when $\rho = 1$, deposits must be fully backed.

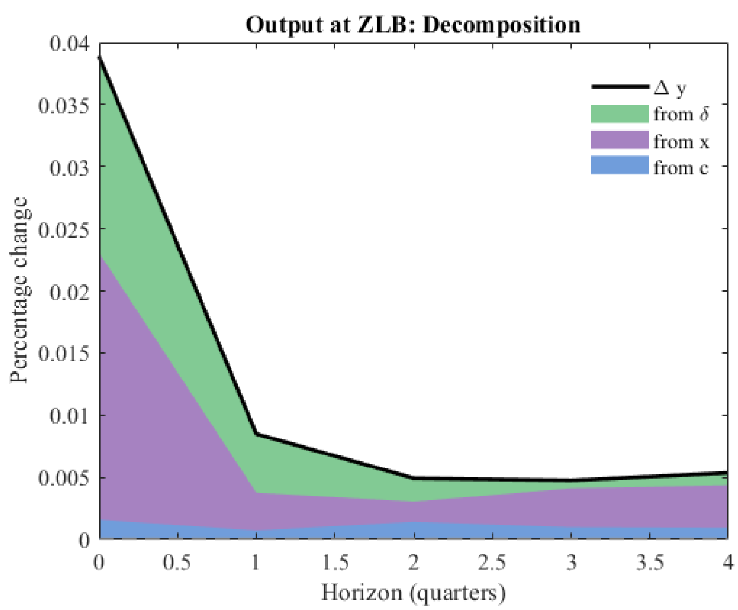


Figure 24: Expansionary FG shock at the ZLB

Here is decompose the response of output to the expansionary FG shock at the ZLB by relative contribution of the relevant variables to the overall change in output.

This final plot above, Figure 24, shows the reason why lending and, in addition, bank balance sheets are of interest when it comes to forward guidance. In wake of the expansionary forward guidance shock at the ZLB, the primary drivers of output's immediate response are bank equity and leverage. Since leverage is constructed as loans/equity and equity increases by more than one-for-one with loans, leverage as a whole falls, enabling banks to further expand credit - hence the jump in lending in figure 6.

After the expansionary shock, the policy rate remains fixed. Nevertheless, the interest rate of loans falls, which reduces marginal costs for firms, which increases production and therefore consumption. However, the financial sector remains the dominant force in driving changes in output.

This elucidates the fact that the primary driver of macro aggregates like output and inflation in response to forward guidance shocks at the ZLB are adjustments made by the financial sector, particularly adjustments of the balance sheets of intermediaries, to these shocks. These shocks loosen bank balance sheets by boosting their equity, lowering their leverage, and thereby enabling them to expand credit, stimulating output as a result. Therefore, the bank balance sheet channel is key in the transmission of forward guidance to the real economy.